Thermochemical Simulation of the Iris Phenolic Liners

Detailed Physics, Implementation and Accepted Results

MIT Rocket Team — Solid Propulsion Simulations

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Exploratory study. Four configurations are considered: sustainer and booster, each with an aluminium or carbon-fibre composite casing. Heat flux, firing duration, grain port and measured recession are unavailable. The user authorized explicit surrogate inputs and sensitivity studies. This report separates native numerical checks from physical calibration. A pilot is not a demonstrated motor lifetime.

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1 Executive summary

No defensible real-motor burnthrough time has been established. The corrected midpoint model has 4 audited terminal results among the four primary cases at this snapshot. Every history ends at its accepted horizon or property-scope stop; no later values are inferred. Six native prescribed-temperature checks verify the rebuilt material library. Historical runs exposed separate restart, internal-substep accounting, nonlinear-tolerance and reaction-temperature issues, documented in the verification sections. Their curves are not used as corrected-model predictions. Booster Al completed the prescribed seven-second heating pulse and cooling through 300 s. Its retained mesh thickness is 6.2706 mm, with a sampled inner-casing maximum of 84.78 degrees Celsius at 142.75 s. Booster CF completed the prescribed seven-second heating pulse and cooling through 300 s. Its retained mesh thickness is 6.2706 mm, with a sampled inner-casing maximum of 114.24 degrees Celsius at 102.75 s. Sustainer Al completed the prescribed seven-second heating pulse and cooling through 127 s. Its retained mesh thickness is 4.7030 mm, with a sampled inner-casing maximum of 120.50 degrees Celsius at 79.00 s. Sustainer CF stopped at its provisional casing-property flag near 34.000 s with 4.7030 mm of retained mesh; its later thermal history is outside this calculation's scope. All four native quarter-step comparisons now pass the casing/scope-timing and retained-thickness targets. Casing peak shifts are at most 0.721 K; sustainer CF property-scope timing changes from 34 to 35 s. Hot-face histories retain larger time-step sensitivity, shown separately; these passes do not establish spatial or surface-temperature convergence. The fresh sustainer Al 160/40-row geometry control has completed its full 127-second history and passed terminal native/restart and independent checks, including both removal events and cooling. Its comparison with the older 80-row case also changes sector geometry and solver tolerance (Section 13). The verified sustainer CF 160/40-row case reaches its provisional casing-property flag at 31.25 s with 4.7030 mm retained. Its combined numerical comparison changes scope timing by 10.71 percent, exceeding the five-percent target. This is not a burnthrough or failure time; 6 further resolution cases remain. Booster Al with 160/40 rows completes 300 s with an inner-casing peak of 87.27 degrees Celsius at 141.25 s and 6.3103 mm retained. Its mixed numerical comparison changes casing peaks by at most 2.711 K and retained thickness by 0.03969 mm; both stated targets pass, while finer mesh/time checks remain necessary. Independent tests also show that the residual-solid cutoff materially affects sustainer CF scope timing. Current native values remain conditional on the sealed 0.1-percent removal rule; tighter-cutoff convergence is investigated separately. No native exhaustion event has been accepted in these scenarios.

The initial twenty-row phenolic mesh is insufficient for recession. Independent refinement and 54 input-sensitivity cases show that missing thermal inputs and the partial-cell removal approximation matter. Under sustained nominal heating, provisional casing-property flags are reached with substantial liner remaining. Those stopping points limit this model's scope; they are not motor lifetimes or structural failure predictions. The immediate objective is to complete the native refined histories and compare their temperatures, mass loss and cooling against independent calculations.

2 Purpose and definition of burnthrough

The primary event is exhaustion of the continuous phenolic barrier, leaving the casing exposed in this two-layer geometry. Pyrolysis conversion alone is not this event: char remains solid and can continue insulating. We report modelled solid mass, retained mesh thickness, whole-cell recession, conversion depth and casing temperatures separately. Retained mesh thickness is not a measurement of physical liner thickness. A numerically accepted exhaustion event would be bracketed by the last intact state and first exhausted state.

The calculation represents an initially intact, locally uniform wall. Cracks, joints, debonding, char spallation and particle erosion are not resolved. Material-temperature flags indicate the scope of surrogate properties, not certified pressure-casing failure. If the event is absent, the result is "not reached through the simulated exposure," with residual thickness, rather than an extrapolated lifetime.

3 Interpreting the temperatures: gas, liner surface and casing

The exposed phenolic is already at thousands of degrees in absolute temperature. The accepted primary runs reach approximately 1480–1488 degrees Celsius (1753–1761 K) at their exposed phenolic surface. The much lower 85–150-degree values discussed in the casing results belong to the metal/composite behind the liner, generally during subsequent heat soak. They are not chamber-gas temperatures or peak exposed-liner temperatures. A cool final point on a 127 or 300 s history is not the maximum during the seven-second heating interval.
Moving radially outward, the temperature locations are

$$\boxed{\text{Gas } T_g \text{ (prescribed)}} \longrightarrow \boxed{\text{Exposed phenolic } T_s \text{ (solved)}}$$

$$\underbrace{\qquad\qquad}_{\text{4.7625 or 6.35 mm initial liner}} \longrightarrow \boxed{\text{Inner casing } T_{ci}} \longrightarrow \boxed{\text{Outer casing } T_{co}}.$$

The gas is a thermal reservoir at 2504 K (2230.85 degrees Celsius), coupled through a finite heat-transfer coefficient. It is not a temperature constraint applied to the solid. T_s is the area-weighted mean on the currently exposed row, whereas T_{ci} in the tables is the maximum nodal temperature at the fixed liner/casing interface. They differ in location and averaging definition. The sector uniformity checks quantify the small axial/angular spreads.

Case	Peak T_s °C	Time of peak s	T_s at 7 s °C	T_{ci} at 7 s °C
Sustainer Al	1479.60	4.1625	1466.03	23.37
Sustainer CF	1479.60	4.1625	1466.03	29.60
Booster Al	1487.71	4.6000	1452.14	22.04
Booster CF	1487.71	4.6000	1452.14	22.34

Table 1. Accepted primary native 80/20-row values. Hot-surface peaks are sampled area-weighted means during heating, not the prescribed gas temperature. Spatial convergence remains in progress.

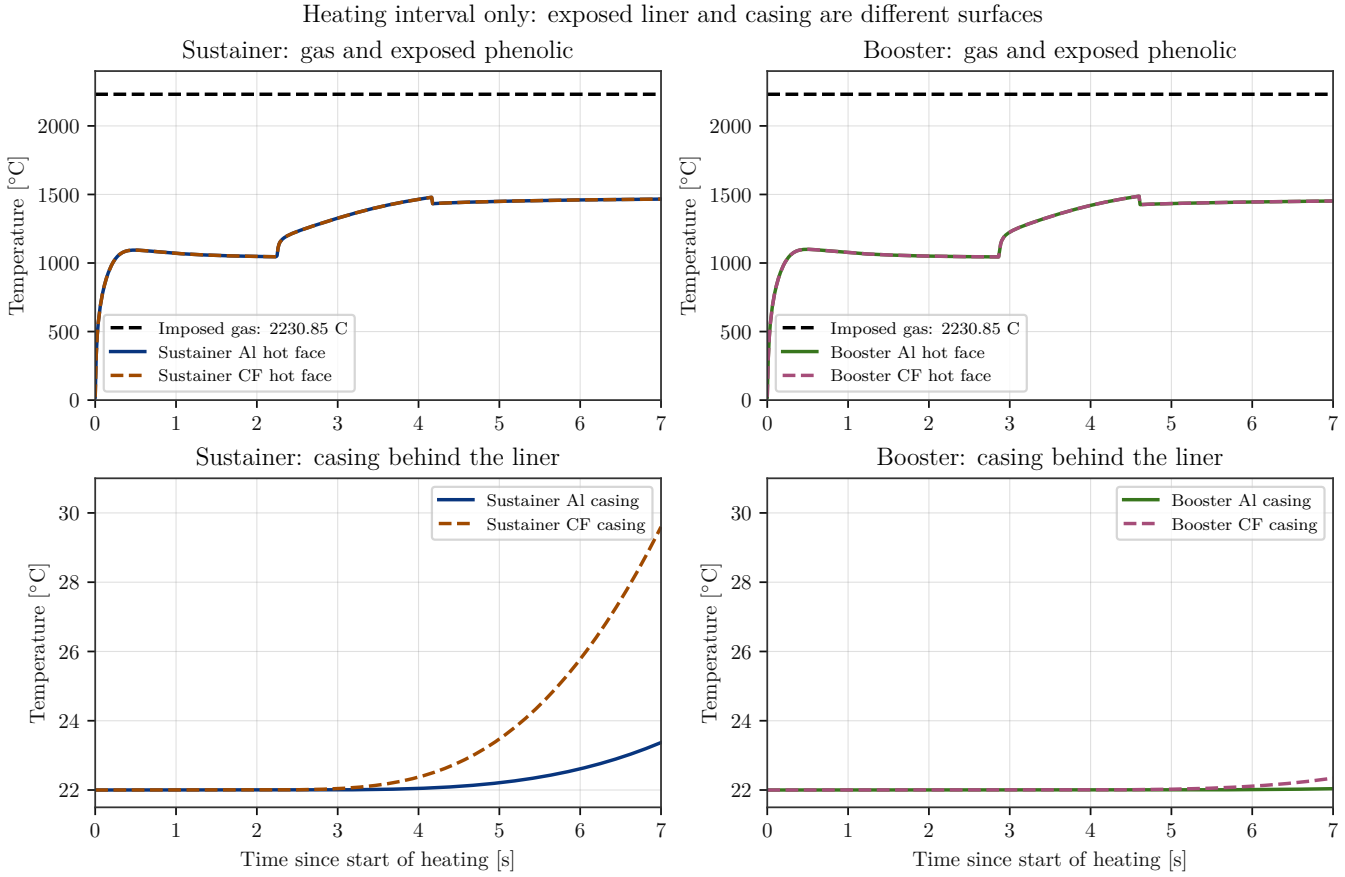


Figure 1. The same heating interval plotted at the exposed phenolic and the liner/casing interface. The lower panels deliberately use a much smaller temperature scale. Al/CF hot-face curves nearly coincide during the short exposure.

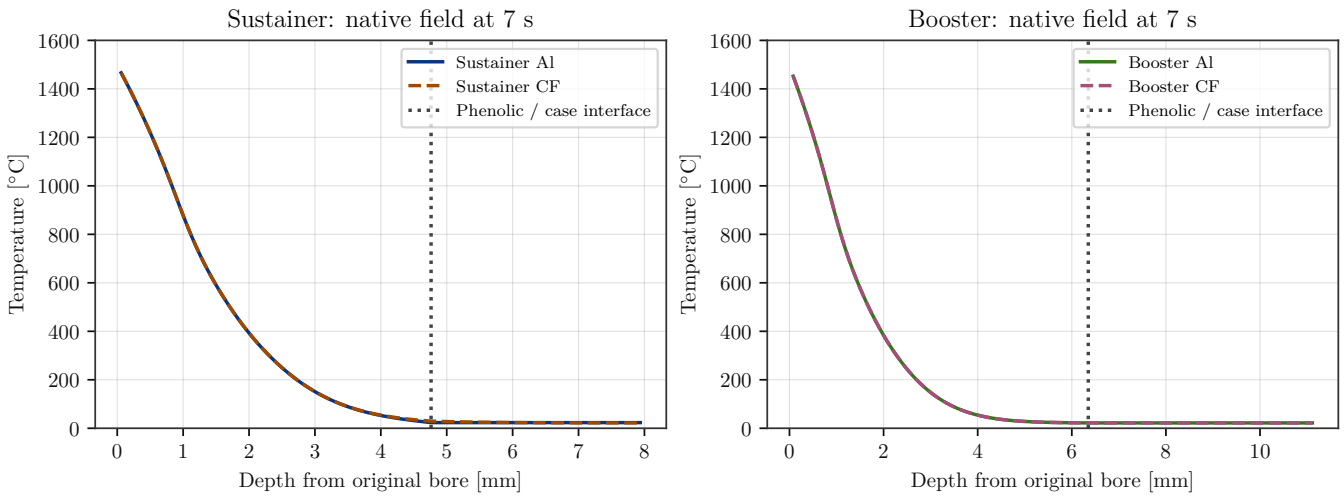


Figure 2. Native radial-station averages at the last heated state. Inactive original-bore nodes are excluded. The steep temperature gradient lies inside the phenolic; the casing remains much cooler during firing and warms later. These are finite-mesh fields, not measured temperatures.

3.1 Why a 2504 K gas does not impose a 2504 K liner

At 7 s, the sustainer hot surface is about 1466 degrees Celsius and its applied film coefficient is about 923 W/(m^2 K). Their product $h_{\text{eff}}(2230.85 - T_s)$ gives approximately 0.706 MW/m^2 of convection into the surface. Booster values give about 0.717 MW/m^2. This was checked against the actual native face heat rates. The small absolute powers in the short-sector records are a consequence of the small area, not a weak heat flux.

Case	Convection inward	Pyrolysis sink	Gas sensible sink	Char cost sink	Capacity rate
	MW/m^2 of exposed hot area, at 7 s				
Sustainer Al	0.7061	0.0550	0.1209	0.2777	0.2524
Booster Al	0.7174	0.0564	0.1253	0.2826	0.2531

Table 2. Whole-domain power terms divided by current hot area. Volumetric sinks and capacity are integrated through the domain; this table is not a local partition at one integration point. Exterior losses and the small discrete residual complete the balance.

Heat is consumed by decomposition and the specified gas/char sinks while the remaining energy heats the solid and conducts outward. A rough initial virgin diffusivity is $k/(\rho c_p) = 0.25/(1250 \times 900) = 2.22 \times 10^{-7}$ m^2/s; its seven-second diffusion length $\sqrt{at}$ is about 1.25 mm, compared with 4.76 or 6.35 mm of initial liner. This is only an order-of-magnitude explanation of the delayed casing response: properties and the thermal front evolve, so it is not a substitute for the nonlinear calculation.

3.2 Comparison with the attached earlier documentation

The September 18 report [1], Table 6 on page 11, reports a hot-surface maximum of 1755.39 degrees Celsius at 6.3856 s while the aluminium maximum remains 22.0012 degrees Celsius. Its Simulation 2 reports a 1644.11-degree maximum at 6.45 s. Thus the earlier documentation also separates a very hot exposed liner from a much cooler casing. Those cases used a different four-layer construction, and Simulations 1 and 2 used the documented oversized radial geometry. Simulation 1 also lacked the later outgassing/consumption terms. They are not same-geometry temperature benchmarks for Iris.

The corrected thin-sector Simulation 3 in that report reached a peak of 1400.50 degrees Celsius and ended at 435.36 degrees after row transfers; those historical quantities use their own deletion rule and do not calibrate the present model. The later September 26 two-layer v0 study is the principal implementation ancestor of Iris. The current source uses its density/char formulation with explicit numerical corrections, not a verbatim reuse of every earlier model.

The present hot face is therefore not accidentally set to approximately 100 degrees Celsius. However, this does not demonstrate that its predicted temperature is physically correct. Actual heating, hot radiation, propellant shielding, tailoff, material properties and char removal remain uncalibrated. The retained hot-surface peaks and row-transfer

jumps are also mesh/time dependent. The current 80-row primary fields should not be called final converged surface temperatures. The refinement and input-sensitivity sections quantify what has been checked without forcing a temperature to match an expectation.

3.3 Completed native time check: the hot face still needs refinement

The completed quarter-step comparisons keep the same 80/20-row geometry and physical inputs. Each passes the native audits and its same-mesh independent check. The separate casing-peak or material-scope timing criteria do not establish a time-converged hot-face history.

Case	T_s at 7 s [°C] Primary	Quarter step	Change K	Maximum matched heating difference [K]
Booster Al	1452.14	1419.62	-32.52	63.09
Booster CF	1452.14	1419.62	-32.52	63.09
Sustainer Al	1466.03	1422.38	-43.66	48.69
Sustainer CF	1466.03	1422.38	-43.66	48.69

Table 3. Native-to-native time-resolution evidence. Matching means the same sampled physical times; a different row-transfer time can change the represented exposed surface. This table does not compare ANSYS with experimental measurements.

For Booster Al, the sampled hot-face peak changes from 1487.71 to 1487.19 degrees Celsius, although its temperature at the end of heating changes more. The largest matched heating-history difference occurs at 4.60000 s. For Booster CF, the sampled hot-face peak changes from 1487.71 to 1487.19 degrees Celsius, although its temperature at the end of heating changes more. The largest matched heating-history difference occurs at 4.60000 s. For Sustainer Al, the sampled hot-face peak changes from 1479.60 to 1479.34 degrees Celsius, although its temperature at the end of heating changes more. The largest matched heating-history difference occurs at 0.01250 s. For Sustainer CF, the sampled hot-face peak changes from 1479.60 to 1479.34 degrees Celsius, although its temperature at the end of heating changes more. The largest matched heating-history difference occurs at 0.01250 s. The remaining refinement must assess surface temperatures and transfer events as well as casing temperatures. The thousands-of-degrees scale is present in both native solutions; the exact hot-face history remains provisional.

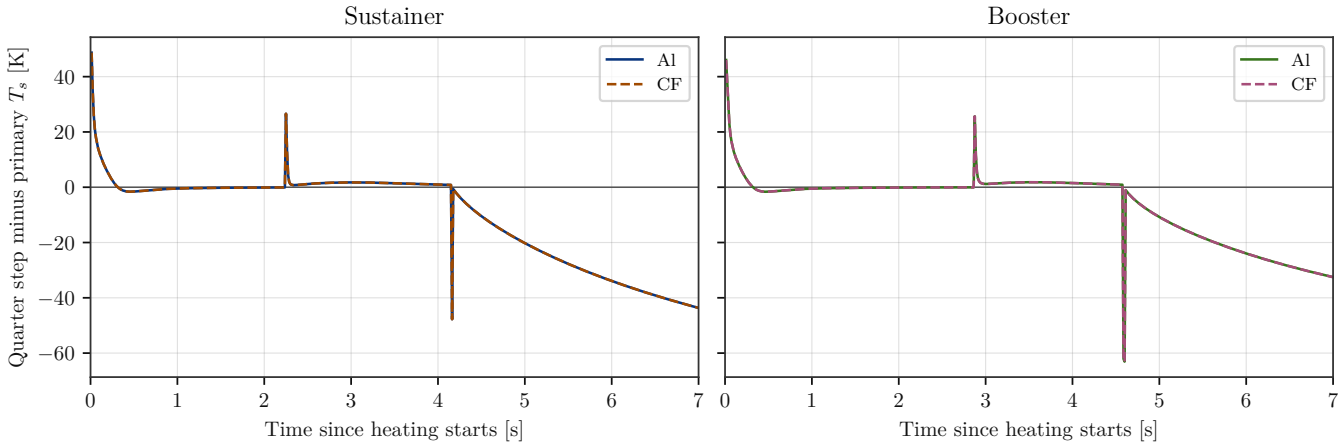


Figure 3. Accepted native hot-face differences at identical sampled times, using the same 80/20-row mesh. Al and CF curves nearly overlap within each motor. Sharp differences accompany start-up and row transfer; a shifted transfer changes which surface is exposed. These curves demonstrate sensitivity to the coupling interval, not experimental error or completed surface-temperature convergence.

4 Geometry and discretization

The supplied propellant radii are provisionally used as liner inner radii. No unreported gap or adhesive layer is inserted. Shared interface nodes impose perfect thermal contact. Actual bond/contact resistance and manufacturing thickness tolerances are unavailable and are not quantified by the material-property perturbations. The user requested short axial and sector slices with about twenty radial cells per material. Each exchanged STEP contains exactly two solids and is reimported independently to verify volumes and dimensions.

Motor	Bore radius	Phenolic	Case	Outside radius
	Dimensions in millimetres			
Sustainer	39.6875	4.7625	3.1750	47.6250
Booster	65.0875	6.3500	4.7625	76.2000

Al and CF versions share the dimensions within each motor. The initial domain is 0.5 mm long and 0.5° wide, with 2 axial and 2 angular divisions. Twenty radial cells in each material give 160 SOLID278 elements and 369 nodes. The axis is global Y. Lateral and axial faces are adiabatic under the uniform-wall assumption. Integrated mass/power scales by 720 to a full ring of the same short length; temperatures and recession do not scale.

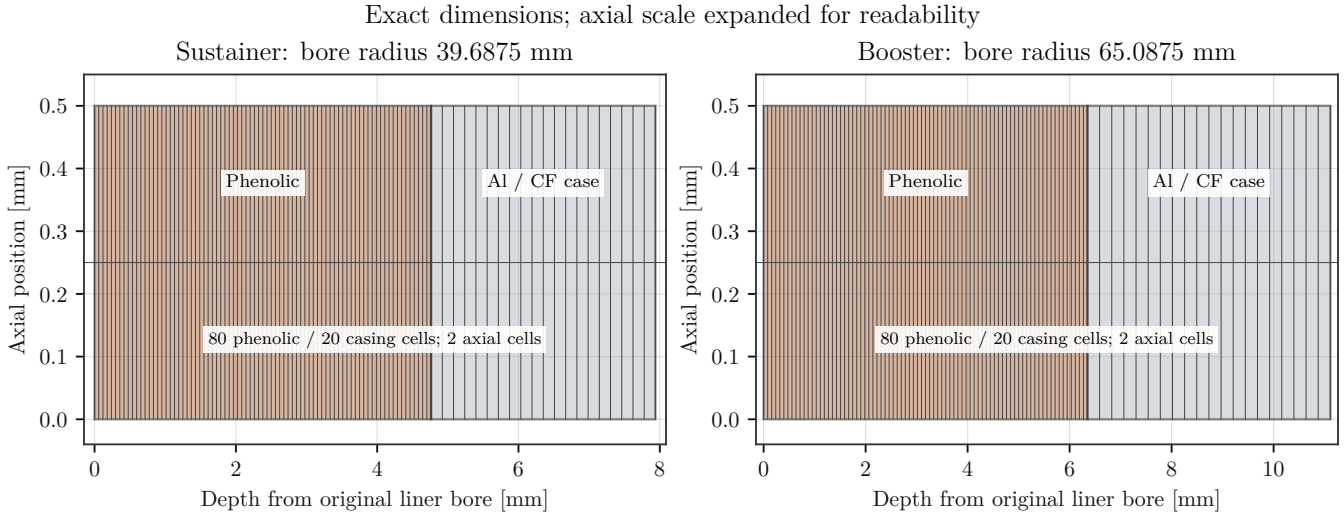


Figure 4. Current primary 80/20-row radial/axial mesh projections. The angular direction has two divisions; axial scale is expanded. These are setup diagrams, not solved contours.

STEP is the exchange geometry. The native structured mesh is generated deterministically from the same dimensions; it is not an automatic mesh imported from STEP. The initial 20-cell phenolic pitches are 0.238125 mm and 0.3175 mm. Independent comparisons at 40, 80 and 160 cells show sensitivity to thermal gradients and cell-removal quantization. Refined native inputs use 80 phenolic and 20 casing rows, giving 400 elements and 909 nodes. Further independent refinement assesses retained partial-cell storage and time accuracy.

5 Materials and thermal environment

5.1 Phenolic and aluminium inherited from the earlier study

The reference virgin and char densities are 1250 and 600 $\mathrm{kg\,m^{-3}}$. The inherited single-step kinetics use $A = 333\,\mathrm{s^{-1}}$, $E_a = 64\,081\,\mathrm{J\,mol^{-1}}$, order one, and an endothermic pyrolysis term of $418\,\mathrm{kJ\,kg^{-1}}$. Thermal tables vary with temperature. Their applicability to the manufactured Iris liner is uncalibrated. Aluminium initially uses $\rho = 2700$, $k = 167$ and $c_p = 896$ in SI units; the exact alloy is unspecified.

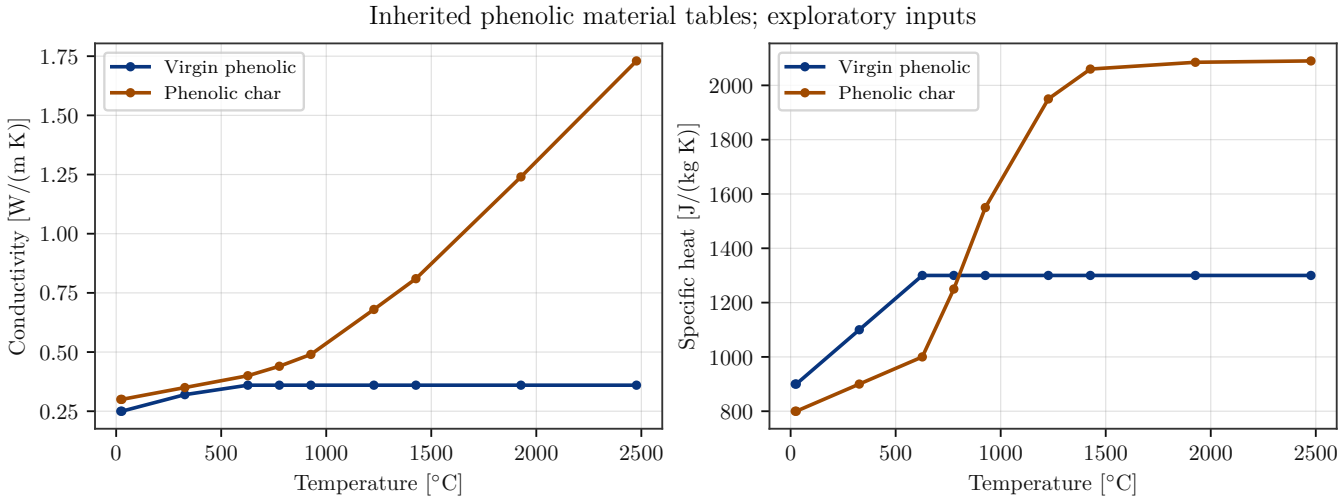


Figure 5. Inherited input tables. Lines interpolate prescribed material data and are not experimental measurements made for Iris.

5.2 Carbon fibre surrogate

The user's ACP fin tutorial supplies $\rho = 1451\,\mathrm{kg\,m^{-3}}$, elastic and thermal-expansion data, but no CF conductivity or heat capacity. Its G10 thermal values are not used as CF properties. The initial woven-laminate surrogate has radial conductivity $0.55\,\mathrm{W\,m^{-1}\,K^{-1}}$ and equal tangential/axial conductivities $2.5\,\mathrm{W\,m^{-1}\,K^{-1}}$, within the woven-composite ranges reported by Joven and colleagues [6]. Local material frames follow the cylinder.

For heat capacity, use the separately identified T700/M21 surrogate $c_p = 2.8773T + 687.31$, with Celsius temperature and SI heat-capacity units [7]. Clamp this input above 200 °C; high-temperature predictions then have a flagged material approximation. The density, conductivity and heat capacity come from different references and do not constitute a measured property set for the user's casing. A 150 °C CF flag and 300 °C Al flag track provisional property scope, not structural failure. NASA's alternative carbon-epoxy data provide another sensitivity reference [8].

5.3 Imposed exposure

The initial gas-side assumption is $T_g = 2504\,\mathrm{K}$, unblown $h_0 = 1000\,\mathrm{W\,m^{-2}\,K^{-1}}$, and nominal pressure 800 psi, following the earlier study and the user's pressure estimate. Pressure is interpreted as absolute in the char-law closure. Initial and external ambient temperature is 22 °C; external convection is $8\,\mathrm{W\,m^{-2}\,K^{-1}}$ and emissivity is 0.25. The initial hot-radiation emissivity is zero, with separate radiation sensitivity scenarios.

The liner is assumed exposed from the start of imposed heating. The actual grain shielding and local exposure history are unknown. The campaign includes native pilots and 7 s baselines, refined native 7 s pulses with cooling, independent 7/20/60 s pulse screens, and a separate independent continuous-exposure branch. None of these pulse lengths is an inferred motor burn duration. Sensitivities vary heat-transfer coefficient, radiation, material properties, pressure and char-consumption parameters; they are not a statistical uncertainty envelope.

5.4 Supplied reaction and inventory envelope

The user's reaction is atom-balanced:

$$3\mathrm{NH_4ClO_4} + 3\mathrm{Al} \longrightarrow \mathrm{Al_2O_3} + \mathrm{AlCl_3} + 3\mathrm{NO} + 6\mathrm{H_2O}.$$

Its reactant mass bookkeeping is approximately 81.32 percent ammonium perchlorate and 18.68 percent aluminium. With constituent densities of 1950 and $2700\,\mathrm{kg\,m^{-3}}$, an ideal additive-volume binary mixture would have density $2056.7\,\mathrm{kg\,m^{-3}}$. The AP density is a published material value [9]; the binary mixture is an explicit assumption, not a measured propellant formulation.

Motor	Volume at 1 m [litres]	Full-cylinder mass [kg]	0.75–1.25 m range [kg]
Sustainer	4.948	10.18	7.63–12.72
Booster	13.309	27.37	20.53–34.22

These are full-cylinder envelope values with no port, binder or voids. Actual mass scales with occupied volume fraction and actual bulk density. The stoichiometric product list is not an equilibrium combustion calculation, and

this inventory cannot establish burn duration, wall heat flux or char kinetics. The inherited gas temperature remains an imposed scenario. No full-length fuel cylinder is meshed.

6 Thermochemical equations and their implementation

6.1 Three distinct processes, and what this model solves

Propellant combustion supplies the chamber environment; phenolic pyrolysis converts the liner into gas and char; surface consumption removes some of that char. These are three different processes. The supplied AP/aluminium reaction is inventory bookkeeping, not the phenolic reaction and not a calculation of the chamber temperature. No Fluent combustion, species-equilibrium, pressure-flow or propellant-regression calculation is being executed. The gas temperature, heat-transfer coefficient and exposure duration are prescribed. Nominal pressure enters the effective char-consumption law only; it is not a structural pressure load.

The native calculation is a nonlinear transient thermal solution in Mechanical APDL 2026 R1.02. The phenolic uses the compiled Fortran `UserMatTh` routine, exported as `USERMATTH`. This is the thermal material interface, not structural `USERMAT`. The casing uses built-in thermal properties. A separate Python controller updates the surface environment and row-removal state between native solves. The equations below describe the executed corrected-midpoint implementation, not every mechanism proposed in the older report.

6.2 Local energy equation, units and mass basis

Let T_C denote temperature in degrees Celsius and $T_K = T_C + 273.15$ its absolute value. Temperature differences and gradients have the same numerical value in kelvins and Celsius. Arrhenius rates and fourth-power radiation always use kelvins. With inward boundary heat flux positive, the implemented solid equation on the current live mesh is

$$\rho_s(T,\alpha)c_p(T,\alpha)\frac{\partial T}{\partial t} = \nabla\cdot\big(\boldsymbol{k}(T,\alpha)\nabla T\big) - \rho_s \Delta H_p \bar{\dot{\alpha}} - Q_g, \tag{1}$$

where ρ_s is in $\mathrm{kg\,m^{-3}}$, c_p in $\mathrm{J\,kg^{-1}K^{-1}}$, conductivity in $\mathrm{W\,m^{-1}K^{-1}}$, and each volumetric source in $\mathrm{W\,m^{-3}}$. Fourier heat flux is $\boldsymbol{q} = -\boldsymbol{k}\nabla T$. Phenolic conductivity is isotropic; the CF casing has a locally oriented conductivity tensor.

The implemented endothermic coefficient is $\Delta H_p = 418000\ \mathrm{J\,kg^{-1}}$. For these purely thermal elements, `hgen` is a source per unit mass; the routine subtracts $\Delta H_p \bar{\dot{\alpha}}$ and the gas sensible sink from its incoming value [2]. Multiplication by the updated apparent density gives the volumetric pyrolysis sink in Equation (1). Thus the numerical mass basis is the current mixture density, not a fixed initial virgin density and not the released-gas mass. Changing that basis would change the model and require new sealed cases.

Conservation limitation. For a reacting, mass-losing control volume, $\partial(\rho u)/\partial t$ does not reduce automatically to $\rho c_p \partial T/\partial t$: density-change terms and the enthalpy carried by escaping mass must be consistently accounted for. Equation (1) is the inherited apparent-capacity closure, with separately prescribed reaction and gas sinks. It is not a derivation of a fully conservative porous-ablation enthalpy equation. Passing the discrete energy audit establishes consistency with this closure. It does not establish that the chosen reaction enthalpy, gas sensible term and removed-cell energy form a calibrated physical enthalpy model. In particular, calorimetric data would need to distinguish the reaction enthalpy from gas sensible enthalpy to avoid double counting.

6.3 Pyrolysis kinetics: derivation and numerical increment

The local irreversible conversion α is zero for virgin phenolic and one for the char residue. The law and constants are

$$\dot{\alpha} = k_r(T_K)(1-\alpha)^n, \qquad k_r(T_K) = A\exp\left(-\frac{E_a}{RT_K}\right), \tag{2}$$

with $A = 333\ \mathrm{s^{-1}}$, $E_a = 64081\ \mathrm{J\,mol^{-1}}$, $R = 8.31446261815324\ \mathrm{J\,mol^{-1}K^{-1}}$ and executed order $n = 1$. These are inherited single-reaction parameters, not a fit to the actual Iris phenolic. The model has no explicit decomposition-product species or multiple resin reactions.

Over one attempted native time increment, the rate constant is frozen at the temporal midpoint. Put $\beta = 1-\alpha$. For $n = 1$,

$$\int_{\beta_n}^{\beta_{n+1}} \frac{d\beta}{\beta} = -k_r(T_{n+1/2})\int_{t_n}^{t_{n+1}} dt, \qquad \boxed{\alpha_{n+1} = 1-(1-\alpha_n)e^{-k_r(T_{n+1/2})\Delta t}}. \tag{3}$$

The source uses the increment-average rate

$$\bar{\dot{\alpha}} = \frac{\alpha_{n+1}-\alpha_n}{\Delta t}. \tag{4}$$

This is exact for a constant rate coefficient during that increment, not exact integration of an arbitrarily changing temperature history. The implemented general-order branch, unused by the nominal configurations, is

$$\beta_{n+1} = \left[\beta_n^{1-n} - (1-n)k_r\Delta t\right]^{1/(1-n)}, \qquad \alpha_{n+1} = 1 - \beta_{n+1}, \quad n \neq 1, \tag{5}$$

with bounds to avoid a nonphysical bracket and to keep $\alpha_n \leq \alpha_{n+1} \leq 1$.

In `UserMatTh`, `Temp` is the current endpoint and `dTemp` is its increment. Consequently the corrected code evaluates

$$T_{K,n+1/2} = \texttt{Temp} - \tfrac{1}{2}\texttt{dTemp} + 273.15. \tag{6}$$

The printed routine in the September report instead adds half the increment. That computes an extrapolated temperature during heating, not the intended midpoint. Six native increasing/decreasing/constant-temperature checks verified the correction; constant-temperature tests alone could not reveal it. Historical curves retain their historical routine and are not silently relabelled.

If $\Delta\alpha > 0.05$, the routine requests a cutback with factor

$$f_{\text{cut}} = \max\left(0.1, \min\left(0.95, \frac{0.05}{\max(\Delta\alpha, 10^{-30})}\right)\right). \tag{7}$$

This guards conversion during a native increment. The production controller additionally requires exactly one native substep per accepted coupling interval. If MAPDL internally subdivides an interval, the present endpoint audit cannot accept it; it stops rather than treating the last internal rate as the rate over the full interval. Smaller prescribed intervals must be prepared in a new, separately verified package.

6.4 Virgin/char mixture and complete property table

At the updated conversion, the material returns

$$X(T,\alpha) = (1-\alpha)X_v(T) + \alpha X_c(T), \qquad X \in \{k, c_p, \rho_s\}. \tag{8}$$

ANSYS interpolates the temperature-dependent input properties before passing them to the routine. In Iris, the reference densities are constant at $\rho_v = 1250$ and $\rho_c = 600\ \mathrm{kg\,m^{-3}}$; only their mixture changes with conversion. These constants follow the later two-layer v0 model. They differ from the temperature-dependent 1250–1150 and 700–600 density tables printed for Simulation 1 in the September 18 report. That distinction matters for gas yield and heat capacity.

T_C °C	T_K K	k_v	k_c W/(m K)	$c_{p,v}$	$c_{p,c}$ J/(kg K)	ρ_v/ρ_c kg/m^3
22.00	295.15	.25	.30	900	800	1250/600
26.85	300	.25	.30	900	800	1250/600
326.85	600	.32	.35	1100	900	1250/600
626.85	900	.36	.40	1300	1000	1250/600
776.85	1050	.36	.44	1300	1250	1250/600
926.85	1200	.36	.49	1300	1550	1250/600
1226.85	1500	.36	.68	1300	1950	1250/600
1426.85	1700	.36	.81	1300	2060	1250/600
1926.85	2200	.36	1.24	1300	2085	1250/600
2476.85	2750	.36	1.73	1300	2090	1250/600

Table 4. Actual nominal Iris `TB,USER` table. The added 22-degree point covers the initial state. The printed code appendix contains all 15 properties at every point.

The interpolation is an engineering mixture rule. It does not model pore geometry, shrinkage, cracking, pressure dependence of conductivity, or a separate fibre/resin microstructure.

6.5 Gas generation, escape and sensible enthalpy

With fixed reference densities and gas-yield multiplier $Y = 1$, conversion produces

$$\dot{\omega}_g''' = Y(\rho_v - \rho_c)\overline{\dot{\alpha}}, \qquad \rho_{g,\text{released}} = Y(\rho_v - \rho_c)\alpha. \tag{9}$$

The multiplier $Y = 1$ is not a 100-percent mass yield: the fully converted gas fraction relative to initial virgin mass is $(1250 - 600)/1250 = 0.52$, leaving 0.48 as char before surface consumption. The gas rate is integrated over all live phenolic elements. For an ideal axisymmetric slice the corresponding surface flux is

$$\dot{m}''_g(r_s) = \frac{1}{r_s}\int_{r_s}^{r_p} \dot{\omega}'''_g(r)\, r\, dr = \frac{\sum_e \dot{\omega}'''_{g,e} V_e}{A_s}. \tag{10}$$

The native controller implements the latter volume/area sum, using the accepted conversion increment. It assumes immediate escape to the exposed surface. It does not solve gas storage, permeability, Darcy velocity, pore pressure or separate gas temperature. The Darcy equations discussed in the earlier report are context for a more complete model, not equations executed here.

The local positive sensible-energy sink is

$$Q_g = f_g \dot{\omega}'''_g c_{p,g} \max(T_{K,n+1/2} - T_{g,\mathrm{ref}}, 0), \qquad \dot{q}_{g,m} = Q_g/\rho_s, \tag{11}$$

where $f_g = 1$, $c_{p,g} = 1600\ \mathrm{J\,kg^{-1}K^{-1}}$ and $T_{g,\mathrm{ref}} = 300$ K. The routine returns $\texttt{hgen} = \texttt{hgen}_{\mathrm{input}} - \Delta H_p \bar{\dot{\alpha}} - \dot{q}_{g,m}$. The mass ledger and thermal sink therefore share the same conversion increment. The local sink approximates enthalpy evacuation; it does not solve the gas enthalpy-advection equation.

6.6 Gas-film blowing and boundary conditions

The gas-film correction uses the previous accepted pyrolysis-gas and surface-consumption rates:

$$B_h^n = \frac{(\dot{M}_g^n + \dot{M}_c^n) c_{p,g}}{A_s^n h_0}, \qquad F(B) = \frac{B}{e^B - 1}, \qquad h_{\mathrm{eff}}^{n+1} = h_0 F(B_h^n). \tag{12}$$

The routine uses $F(B) = 1 - B/2 + B^2/12$ for very small B, recovering h_0 at zero blowing. This closure represents resistance to heat transfer by outward gas flow. Treating consumed-char mass as additional blowing flow is an effective assumption, not a species-resolved surface-product mass balance.

The hot-boundary heat terms, positive into the solid, are

$$q''_{\mathrm{conv}} = h_{\mathrm{eff}}(T_g - T_s), \tag{13}$$

$$q''_{\mathrm{rad,h}} = \epsilon_h \sigma (T_{g,\mathrm{rad,K}}^4 - T_{s,K}^4), \tag{14}$$

$$q''_{\mathrm{applied}} = q''_{\mathrm{conv}} + q''_{\mathrm{rad,h}} - H_{\mathrm{eff}} \dot{m}''_{c,\mathrm{applied}}. \tag{15}$$

Here $\sigma = 5.670374419 \times 10^{-8}\ \mathrm{W\,m^{-2}K^{-4}}$. The baseline uses $T_g = 2504$ K $= 2230.85\,^{\circ}$C, $h_0 = 1000\ \mathrm{W\,m^{-2}K^{-1}}$ and $\epsilon_h = 0$. Thus it imposes a hot gas reservoir, not a prescribed wall temperature. The initial unblown convective flux is $1000(2230.85 - 22) = 2.20885\ \mathrm{MW\,m^{-2}}$ and falls as the wall heats and blowing develops.

Convection is imposed through `SFE` and depends on the solved wall temperature within the increment. Radiation and char power are computed explicitly from the preceding accepted state and distributed as equal nodal `F,HEAT` contributions on each four-node face. Pyrolysis and gas sensible cooling act inside `UserMatTh`; they are not subtracted a second time in the boundary load.

The outside boundary uses $h_o = 8\ \mathrm{W\,m^{-2}K^{-1}}$, $\epsilon_o = 0.25$ and $T_\infty = 22\,^{\circ}$C. Its inward radiation term is $\epsilon_o \sigma (T_{\infty,K}^4 - T_{o,K}^4)$. End and angular faces are adiabatic. Shared liner/case nodes impose continuity of temperature and normal conductive heat flow; there is no adhesive volume or contact resistance.

For the primary scenarios, hot exposure ends at exactly 7 s. Thereafter the inner boundary is set to ambient 22 degrees Celsius with $h = 8$, hot radiation is zero and surface char consumption is disabled. Pyrolysis and its gas sensible sink can continue in hot material. This is a specified ambient-cooling experiment; real hot-gas tailoff is unknown. A 127 or 300 s plot is therefore not a 127 or 300 s firing.

6.7 Pressure-dependent char law and available-energy cap

Only char formed at the exposed row can be consumed. The effective kinetic and transport channels are

$$j_{\mathrm{kin}} = j_{\mathrm{ref}} \exp\left[-\frac{E_c}{R}\left(\frac{1}{T_{s,K}} - \frac{1}{T_{\mathrm{ref}}}\right)\right]\left(a_r \frac{p_c}{p_{\mathrm{ref}}}\right)^m, \tag{16}$$

$$j_{\mathrm{tr}} = j_{\mathrm{cap}} a_r \frac{p_c}{p_{\mathrm{ref}}}, \qquad j_c = \left(j_{\mathrm{kin}}^{-1} + j_{\mathrm{tr}}^{-1}\right)^{-1}. \tag{17}$$

The nominal constants are $j_{\mathrm{ref}} = 0.01$ and $j_{\mathrm{cap}} = 0.05\ \mathrm{kg\,m^{-2}s^{-1}}$, $E_c = 100000\ \mathrm{J\,mol^{-1}}$, $T_{\mathrm{ref}} = 1500$ K, $m = 0.5$, $a_r = 1$ and $p_{\mathrm{ref}} = 100000$ Pa. The imposed $p_c = 800$ psi is 5.5158058 MPa, interpreted as absolute; hence $p_c/p_{\mathrm{ref}} \simeq$

55.158. The transport cap is scaled by this ratio: 0.05 is the reference cap, not the actual cap at 800 psi. No measured oxidant mole fractions are supplied. The configurable species branch is unused by the nominal cases.

For one exposed face f of area A_f and cell volume V_f, the actual applied rate is the minimum of four restrictions evaluated from the preceding accepted state:

$$j_{c,f}^{n+1} = \min\left\{ \alpha_f^n j_c(T_f^n, p_c), \quad \frac{\max(q_{\mathrm{conv,f}}^n + q_{\mathrm{rad,h,f}}^n - q_{\mathrm{cond,f}}^n, 0)}{H_{\mathrm{eff}}}, \quad \frac{\max(\alpha_f^n \rho_c V_f - M_{c,f}^n, 0)}{A_f \Delta t}, \quad \frac{0.2 \rho_c V_f}{A_f \Delta t} \right\}. \tag{18}$$

Here $M_{c,f}^n$ is previously consumed char, $q_{\mathrm{cond,f}}^n$ is the last accepted exposed-cell radial-flux estimate, and $H_{\mathrm{eff}} = 35\ \mathrm{MJ\,kg^{-1}}$ is a positive effective removal cost. The first term includes the surface conversion multiplier actually used in the code. The other terms prevent consumption without available surface energy, without produced char, or faster than 20 percent of the reference char inventory per interval. The energy cap is a lagged estimate, not an independently solved surface balance. The controller checks the actual converged global balance afterward. Stronger-heating cases require smaller coupling intervals, as the resolution studies document.

6.8 Solid mass, removal rule and the moving boundary

For a live phenolic cell, the controller's physical-mass ledger is

$$M_{s,e} = \big[\rho_v - (\rho_v - \rho_c)\alpha_e\big] V_e - M_{c,e}. \tag{19}$$

It retains cumulative gas, consumed char and the residual solid ejected at each removal event. On a fixed slice, the ledger satisfies

$$M_0 = M_{s,\mathrm{remaining}} + M_{g,\mathrm{cumulative}} + M_{c,\mathrm{cumulative}} + M_{\mathrm{residual,ejected}}. \tag{20}$$

An entire exposed radial row is scheduled for deactivation once its remaining solid mass is at most $f_{\mathrm{res}} = 0.001$ times that row's initial virgin mass. The leftover mass is explicitly entered in the ejected-residual ledger. At the next solve the previous exposed faces are unloaded, the removed prefix is deactivated with `EKILL`, and convection/radiation/char power move to the next live row. The reported scheduling time and first transferred-load time therefore differ by one coupling interval.

For uniform radial pitch Δr, the reported whole-row recession is $s = N_{\mathrm{scheduled}} \Delta r$ and retained mesh thickness is $L_p - s$. This is distinct from pyrolysis depth, continuous physical recession and the older report's integrated energy-based virtual recession. The code is not using $\alpha = 0.99$ alone as its deletion condition.

Two limitations remain even if the mass ledger closes. First, consumed surface-char mass is subtracted in the ledger but is not removed from the live element's conductivity or heat capacity until the whole row is killed. Second, an exactly zero residual cutoff is not a finite-time physical target for the first-order reaction: available-char capping leaves $M_{s,e} \geq \rho_v(1-\alpha_e)V_e$, and $1-\alpha$ stays positive for finite integrated reaction exposure. The finite cutoff represents unresolved residual ejection or loss of continuity. Its sensitivity must remain visible; it cannot be calibrated by numerical convergence alone. The energy associated with a killed row is logged as a quadrature diagnostic, not as a conservative enthalpy remap or an additional ablation sink.

6.9 Weak form and time-discrete thermal equations

Multiply Equation (1) by a test function N_i and integrate conduction by parts. With $q''_{\mathrm{in}} = -\boldsymbol{q} \cdot \boldsymbol{n}$,

$$\int_{V_l} N_i \rho_s c_p \dot{T}\, dV + \int_{V_l} \nabla N_i \cdot \boldsymbol{k} \nabla T\, dV = \int_{\partial V_l} N_i q''_{\mathrm{in}}\, dS - \int_{V_l} N_i \big(\rho_s \Delta H_p \dot{\overline{\alpha}} + Q_g\big)\, dV. \tag{21}$$

Here V_l is the live solid domain. Using $T_h = \sum_j N_j T_j$ gives the capacity and conduction matrices

$$C_{ij} = \int_{V_l} N_i \rho_s c_p N_j\, dV, \qquad K_{ij} = \int_{V_l} \nabla N_i \cdot \boldsymbol{k} \nabla N_j\, dV. \tag{22}$$

The nominal inputs explicitly set `KEYOPT(15)=1` on SOLID278, selecting diagonal capacity [3]. The independent radial reference uses the corresponding row-summed matrix, $C_{ii}^L = \sum_j C_{ij}$. With backward Euler time integration and convection assembled into H, the equation solved over an accepted interval has the form

$$C^L(T_{n+1}, \alpha_{n+1}) \frac{\boldsymbol{T}_{n+1} - \boldsymbol{T}_n}{\Delta t} + (K + H)\boldsymbol{T}_{n+1} = \boldsymbol{f}_{hT_g} + \boldsymbol{f}_{\mathrm{explicit}} - \boldsymbol{f}_p - \boldsymbol{f}_g. \tag{23}$$

Temperature-dependent properties, conversion, source terms and boundary temperatures make this nonlinear. The full transient/full nonlinear settings are `TRNOPT,FULL`, `THOPT,FULL`, `NROPT,FULL`; `TINTP,,,1` selects the time-weighting

used here. Native logs show the sparse direct solver. Maximum equilibrium iterations are 60, and corrected cases set the heat-flow convergence tolerance to 10^{-5} with a 10^{-6} minimum reference.

The custom routine supplies $\partial \boldsymbol{q}/\partial \nabla T = -kI$ and `dudt` $= c_p$, but sets `dfdt` and `dudg` to zero. It therefore does not return every temperature/history derivative of this nonlinear constitutive law as a fully consistent tangent. This is an implementation limitation; the native tolerance and time-step checks are important even when the analytic conversion update is used.

6.10 Returned states and the coupling sequence

State	Stored quantity	Units
SVAR1	Updated conversion α	dimensionless
SVAR2	Increment-average $\overline{\dot{\alpha}}$	s^{-1}
SVAR3	Gas source $\dot{\omega}_g'''$	$\mathrm{kg/(m^3\ s)}$
SVAR4	Cumulative released-gas density	$\mathrm{kg/m^3}$
SVAR5	Positive gas sensible sink Q_g/ρ_s	W/kg

Table 5. The five state variables in the current compiled routine. They belong only to phenolic elements; casing state-variable entries are not interpreted as phenolic data.

Every accepted interval follows this sequence:

1. Read the last accepted temperatures, conversion, mass ledgers, removed-row count and pinned input hashes. Confirm the storage heartbeat, scope and stopping conditions.
2. Choose the interval without straddling the 7 s switch; compute blowing, radiation and char consumption from the last accepted state. Remove stale loads and replay *every* already removed row after multiframe restart.
3. Check the live-element count, apply convection and nodal heat loads, set the new time and solve in MAPDL. Fortran updates each integration-point reaction state during the nonlinear thermal solution.
4. Export nodal temperatures, phenolic state fields, conductive flux, element capacity rates, face convection rates, and actual load-step/substep identity.
5. Check finite/complete fields, increasing time, nondecreasing bounded conversion, available char, mass closure, actual hot-face power and thermal balance. Reject missing data, unexpected live elements or multiple native substeps.
6. Schedule a depleted row, save mass/energy diagnostics, and stop at the prescribed horizon, a property-scope flag or exhaustion. Atomically save the accepted controller state and two alternating hash-verified native restart copies before the next interval.

The primary intervals are 0.0125 s during heating, 0.0625 s during the first cooling second and 0.25 s afterward. This gives 560 heating solves, followed by 16 early-cooling solves and the later cooling solves. The quarter-step and finer spatial campaigns are listed separately; an analytic local reaction formula does not remove the need for these global time-resolution checks.

6.11 What the numerical audits actually test

The native energy audit sums the exported capacity rate P_{cap}, outward convection $P_{\mathrm{conv,out}}$, native radiation $P_{\mathrm{rad,out}}$ and heat-flux channel P_{hfx}. It independently reconstructs the negative body source $P_{\mathrm{gen}} = -(P_p + P_g)$ and sums the applied nodal power P_N. The residual is

$$R_E = P_{\mathrm{cap}} + P_{\mathrm{conv,out}} + P_{\mathrm{rad,out}} - P_{\mathrm{hfx}} - P_{\mathrm{gen}} - P_N. \tag{24}$$

The normalization is the maximum of $|P_{\mathrm{cap}}|$, $|P_{\mathrm{conv,out}}|+|P_{\mathrm{rad,out}}|$, $|P_{\mathrm{hfx}}|+|P_{\mathrm{gen}}|+|P_N|$, and 10^{-6} W. The acceptance gate is $|R_E|/P_{\mathrm{scale}} \le 0.05$. Mass closure must be within $\max(10^{-13}\ \mathrm{kg}, 10^{-6}M_0)$. Actual native hot-face convection must match $\sum_f A_f h_f (T_{g,f} - T_{s,f}^{n+1})$ within 2 percent.

For this user-material path, the native aggregate generation and convection channels did not reproduce the applied terms. They are retained as diagnostics rather than used unquestioningly. The audit instead reconstructs the material sinks from accepted conversion/SVAR5 and uses native individual-face convection rates (hot face 5, outer face 3). The code appendix prints the exact extraction and balance expressions. Numerical agreement with the independent radial model tests implementation; it does not supply independent experimental evidence for the assumed physics.

6.12 Build, loading and reproducibility

The corrected source was compiled in an isolated directory using Intel Fortran Classic 2021.9.0 from the installed oneAPI 2023.1 environment, Visual Studio 2022 tools and the ANSYS v261 customization libraries. Compilation and linking returned zero, and export inspection found `USERMATTH`. The source, DLL and configuration hashes are pinned in each sealed runtime. The installed Windows x64 solver and DLL execute in the Windows ARM64 guest; this is a native MAPDL calculation, distinct from the separate host radial Python model.

The launcher sets `ANS_USER_PATH` and `ANS_USER_PATH_261` to the runtime directory. It uses the prebuilt DLL; it does not rebuild a user material at every solve. The archived primary command used one SMP core: `ANSYS261.exe -b -smp -np 1 -p ansys -j file`. Continuing cases now use `-smp -np 10`, with one serial solve at a time, as documented in the execution section. The input/output paths select the sealed run. The printed source appendix contains the build commands, material definition, complete Fortran routine, MAPDL driver/resume/export commands and a complete representative mesh deck. These are archival listings, not an instruction to start another solver beside the active serial queue.

6.13 Recorded native warnings

The corrected pilot exporter retrieves only defined live-material ranges. The earlier three undefined-SVAR warnings per step were removed without changing physical equations. The native user-material solve retains the standard warning that the Ansys-supplied USERCK routine is used; its banner separately confirms a user-defined thermal material. This known warning is recorded, not silently counted as zero warnings. Longer runs also report elapsed-time/CPU-time performance warnings; these are retained in the validation records.

Both corrected CF primary runs also record a material-table warning at 21.9999998 degrees Celsius, approximately 2×10^{-7} K below the initial 22-degree table endpoint. The casing conductivity is constant in that interval. This negligible numerical undershoot is retained in the audit; it is distinct from the controller's explicit 150-degree material-scope stop. ANSYS ceased its own repeated temperature-range checking after the warning, while the independent controller scope checks continued at every accepted interval.

The completed booster CF quarter-step successor retains two restart housekeeping warnings: the temporary control files did not yet exist when the resume driver initially attempted to delete them. The normal controller then regenerated them before use. Its final log also recommends DMP instead of ten-core SMP as generic performance advice; no isolated mode benchmark has been performed. Both native logs completed with zero errors, and full terminal state, lineage, balance, topology and independent-reference checks passed. The completed booster Al 160-row case likewise records generic elapsed-time and SMP/DMP performance advisories and finishes with zero solver errors. These messages do not measure the benefit of changing execution mode; the verified ten-core SMP settings are retained. Geometry-control findings and the continuation status are documented in Section 13.

7 Execution and storage controls

The running VM has 14 virtual CPUs and approximately 112 GiB RAM. The repository is accessible through a dedicated I: mapping. A native no-solve probe completed with a successful license checkout and zero warnings/errors. Each case receives a separate sealed runtime; unchanged source hashes and expected inventories are checked before launching.

Current production jobs execute in guest-local `C:\IrisStudy\runs`, with compact accepted checkpoints mirrored to the Mac and complete native files copied back at phase exit. The primary meshes have 400 elements, versus 152000 in the September 26 v0 study: 380 times fewer elements. Axial divisions fall from 250 to 2, while radial divisions fall from 240/64 to 80/20. The sustainer phenolic pitch is therefore three times coarser than the old 19.84375-micrometre pitch; booster pitch is four times coarser. The refinement program is necessary before claiming comparable surface/recession accuracy. One SMP process and guest-local files also avoid much distributed/shared-folder overhead. These changes explain the runtime scale, but are not an isolated performance benchmark or proof of equal accuracy. The shorter slice relies on the locally uniform heating assumption and does not resolve motor ends, joints or axial hot spots.

On November 4 the user requested ten cores. The continuing and remaining native cases now use `-smp -np 10`: one shared-memory process with ten threads, confirmed in the actual solver log. The four primary results and the first three time refinements were computed on one core. The booster CF quarter-step case was cleanly paused at accepted index 1117, $t = 3.490625$ s, and continued in a separate sealed successor with preserved native/controller backups and immutable checkpoint lineage. A short one-core reference and the ten-core continuation agree over 32 resumed steps through 3.590625 s: maximum nodal difference 1.14×10^{-12} K, with identical conversion, mass and integrated energy. No row deletion or cooling transition occurs in this verification interval; all subsequent acceptance checks remain mandatory. Physics, mesh, material DLL, time steps and audit tolerances are unchanged. Only one case runs at a time. The continuing resolution campaign and five deferred controls use ten-core successor packages.

This core count is not a measured tenfold speedup: the small meshes, serial coupling and file transfers also determine runtime. The primary source appendix retains the exact historical primary inputs; the execution change is recorded in `validation/parallel_restart_comparison.json`.

The user's storage floors are 400 GB decimal free on the Mac disk and 10 GB free on Windows C:. A host watcher checks the Mac every five seconds and the guest disk about every thirty seconds. The controller also requires a fresh storage heartbeat. On a threshold crossing, new work stops, a stop latch is written and the active solver is asked to terminate cleanly. Interrupted states are not published as accepted data.

Native results/restarts remain in ignored storage. Major setup and report milestones are pushed to GitHub after verification; staged files and new Git objects at or above 100 MB are rejected. Reports freeze accepted state indices and source hashes. Recovery preserves the failed run and matches native restart state with controller state before continuation.

8 Accepted results at this edition

This snapshot was frozen at November 05, 2026 at 05:43 UTC-05:00. The table and primary histories use corrected midpoint cases only, with tighter heat-flow convergence and strict single-substep accounting. Historical results remain in their explicitly labelled verification sections. Longer runs are displayed only as their states are accepted; queued cases do not acquire predicted ANSYS values. These results are provisional until spatial and temporal comparisons are complete. No burnthrough event has been accepted. The table gives hot-face temperature at the listed endpoint, after cooling; heating maxima are in Section 3.

Run	Time [s]	Endpoint T_s [°C]	Remaining [mm]	Energy error [%]
Booster Al (n=80, midpoint)	300.0000	72.59	6.27062	0.0002
Booster CF (n=80, midpoint)	300.0000	91.49	6.27062	0.0004
Sustainer Al (n=80, midpoint)	127.0000	110.61	4.70297	0.0002
Sustainer CF (n=80, midpoint)	34.0000	249.54	4.70297	0.0012

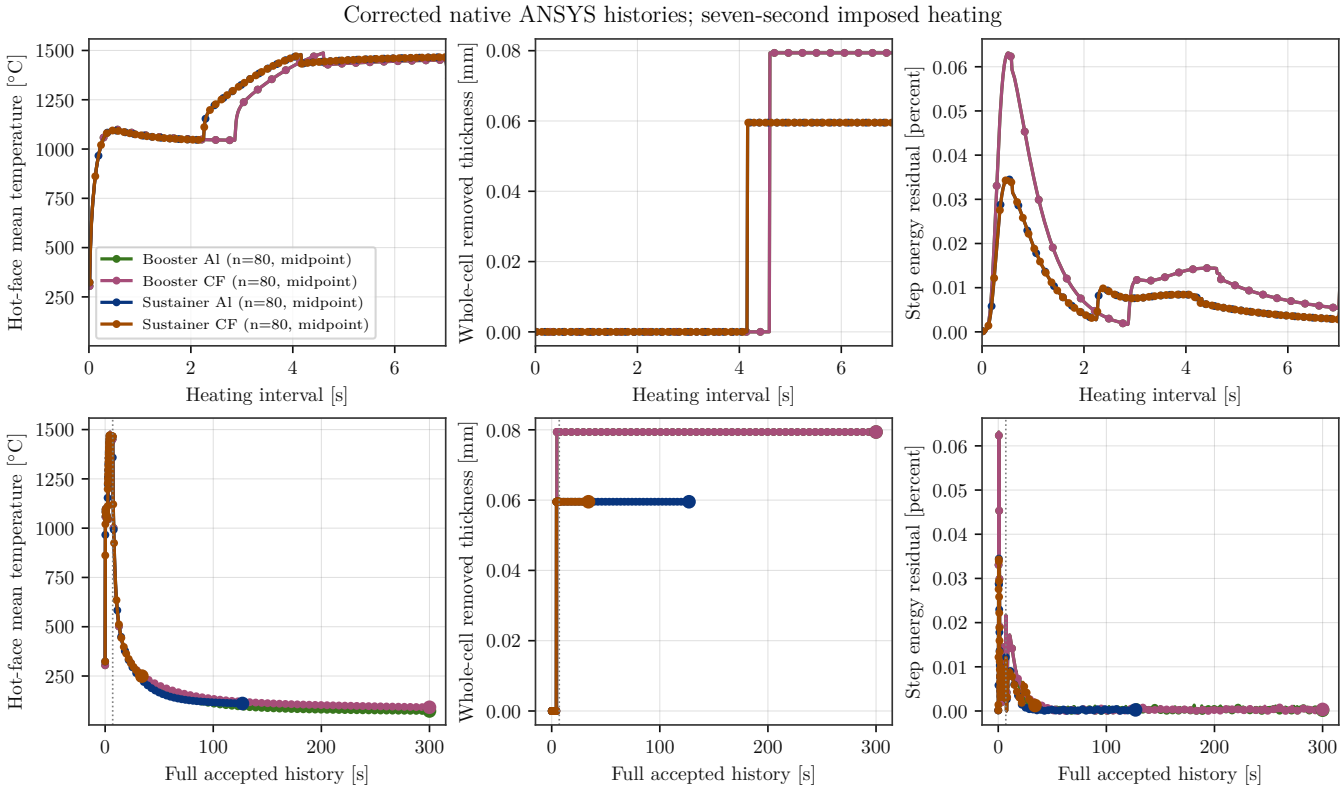


Figure 6. Accepted native histories. No extrapolated future states are included.

9 Verification findings and current limitations

The native steady-conduction benchmark used the same sector mesh, convection face numbers and CF material frames with constant properties and reactions disabled. All four configurations matched the analytical two-layer cylinder within 0.013 K; the acceptance target was 0.1 K. This checks geometry, thermal resistance and orthotropic axes, not the reaction law. The deliberately artificial steady temperatures in that benchmark are not motor predictions.

Manufactured storage faults verified both free-space floors, the latched stop and the correct native abort filenames. A communication fault is retried as a read-only query; it is not reported as a disk threshold crossing. A guest-local pilot reproduced every shared-drive pilot nodal temperature exactly. It offers an alternate execution location while mirroring accepted results to the Mac. The controlled checkpoint restart matched an uninterrupted native run at 1.1 s, including conversion state and energy integrals. The maximum nodal temperature difference was 0 K.

The independent cylindrical finite-element model uses a separate radial assembly, two Gauss points per cell, backward Euler time integration and row-summed heat capacity. It uses the intended midpoint reaction temperature. Early comparisons with the inherited native routine differed by approximately 0.70 K at 0.1 s and 0.97 K at 1.0 s, but did not expose its temperature-increment convention error. The later diagnostic checks below address that distinction. Numerical agreement is not physical calibration.

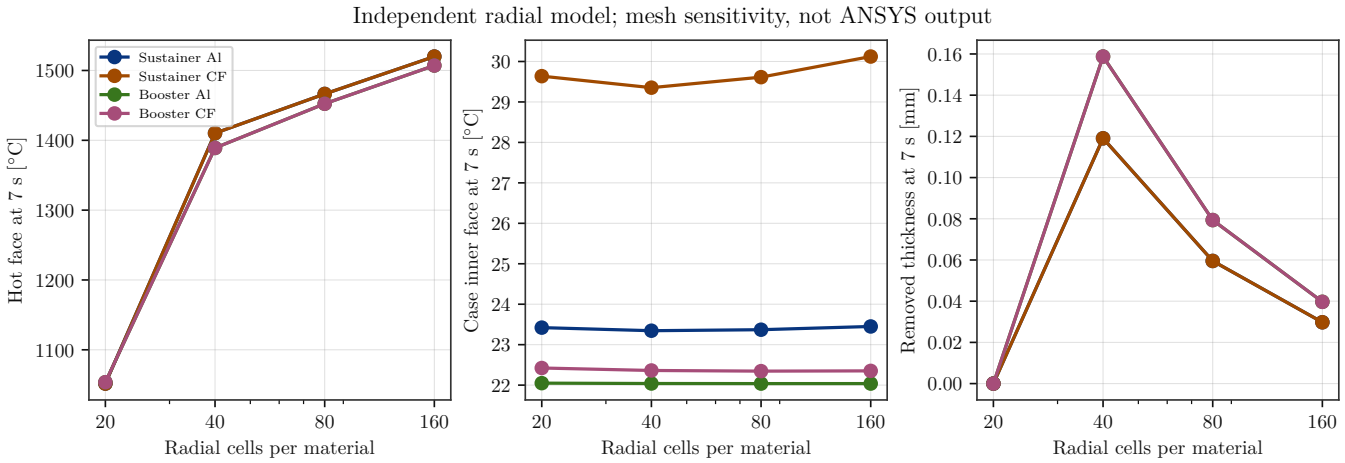


Figure 7. Independent seven-second radial mesh checks at a 0.0125 s coupling step. Coincident Al/CF hot-face and recession curves overlap. A cell-removal jump is not a smooth estimate of recession.

The twenty-cell mesh does not resolve recession sufficiently. The independent sustainer hot face at 7 s changes from about 1052 to 1410, 1466 and 1520 degrees Celsius at 20, 40, 80 and 160 radial cells. Its corresponding whole-cell removed thickness is 0, 0.1191, 0.0595 and 0.0298 mm. Case temperatures are much less sensitive over this short exposure. These comparisons justify refinement; they do not demonstrate a converged burnthrough time. Halving the 80-cell time step also changes hot-face temperature, so mesh and time accuracy remain separate checks.

Numerical long-exposure stress tests were also performed to expose weaknesses in the inherited removal scheme. They showed substantial mesh/time-step dependence, while the casing had already left its provisional property range. Those apparent exhaustion times are explicitly rejected as motor-lifetime results. The retained partial-cell thermal mass and lagged surface coupling require further assessment. Physically interpreted continuous-exposure runs now stop at the first casing property-range crossing. That crossing indicates a limit of this model, not an experimentally established structural failure.

10 Casing temperatures in accepted native states

The maxima below apply only through each accepted time shown. The listed times locate sampled maxima within that history; a material-scope stop cannot locate a later thermal peak. Inner and outer casing temperatures are kept separate because thermal resistance can delay outside heating while the liner-facing side becomes hotter.

Case	Through [s]	Inner max [°C]	At [s]	Outer max [°C]	At [s]
Booster Al (n=80, midpoint)	300.000	84.784	142.75	84.765	142.75
Booster CF (n=80, midpoint)	300.000	114.243	102.75	105.131	144.25
Sustainer Al (n=80, midpoint)	127.000	120.497	79.00	120.477	79.25
Sustainer CF (n=80, midpoint)	34.000	150.063	34.00	112.879	34.00

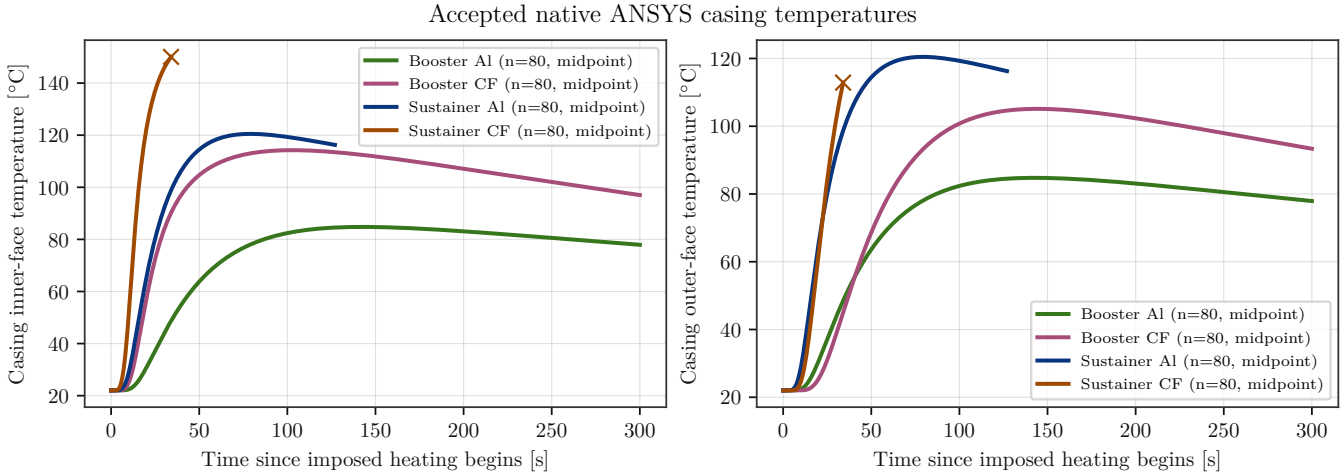


Figure 8. Native accepted casing-temperature histories. An x marks a provisional material-scope stop, not predicted failure. These are thermal calculations, not pressure-casing strength or qualification results.

10.1 Mass bookkeeping and retained partial cells

The next table uses fractions of the initial liner mass at each run's final accepted time above. Gas and consumed char are separate cumulative losses. The last column measures consumed char that remains represented thermally until its cell is removed; it is a numerical storage discrepancy, not extra physical material.

Case	Solid left [%]	Gas loss [%]	Char loss [%]	Retained in FE [%]
Booster Al (n=80, midpoint)	89.404	9.854	0.741	0.169
Booster CF (n=80, midpoint)	89.385	9.873	0.741	0.169
Sustainer Al (n=80, midpoint)	85.767	13.417	0.815	0.249
Sustainer CF (n=80, midpoint)	85.727	13.457	0.815	0.249

A zero whole-cell recession does not mean zero mass loss. Conversely, pyrolysis conversion alone is not a hole through the liner. Refinement is needed to bound the effect of the retained partial-cell storage on temperature and recession.

10.2 Accumulated numerical balance checks

The signed residual below checks accumulated bias; the absolute residual sum avoids cancellation of positive and negative step errors. Both are normalized by the time integral of the step balance scale. The independent mass check recomputes remaining solid plus gas, consumed char and expelled residual against the initial geometric inventory.

Case	Signed energy [%]	Absolute energy [%]	Mass relative error
Booster Al (n=80, midpoint)	0.0134	0.0142	3.2e-15
Booster CF (n=80, midpoint)	0.0132	0.0140	2.9e-15
Sustainer Al (n=80, midpoint)	0.0084	0.0088	3.3e-15
Sustainer CF (n=80, midpoint)	0.0086	0.0091	3.3e-15

The scale is $\max(|\dot{U}_{\mathrm{FE}}|, |Q_{\mathrm{conv}}|, |Q_{\mathrm{body}}| + |Q_{\mathrm{nodal}}|, 10^{-6}\,\mathrm{W})$ integrated over time. These residuals check the solved equations. They are not an independent physical enthalpy balance for departing material. Energy assigned to removed cells remains a separate diagnostic because the inherited capacity law is not a conservative enthalpy remap.

10.3 Corrected native temperature fields

Each map uses accepted corrected-midpoint ANSYS checkpoints only. The left panel resolves the imposed seven-second heating interval; the right panel shows the full accepted history, including heat soak and cooling. The dashed line marks the casing interface, while cyan marks the depth already removed during the solve. Inactive nodes are masked. Temperatures are radial-station averages of the small uniform sector; these are computed fields, not measurements.

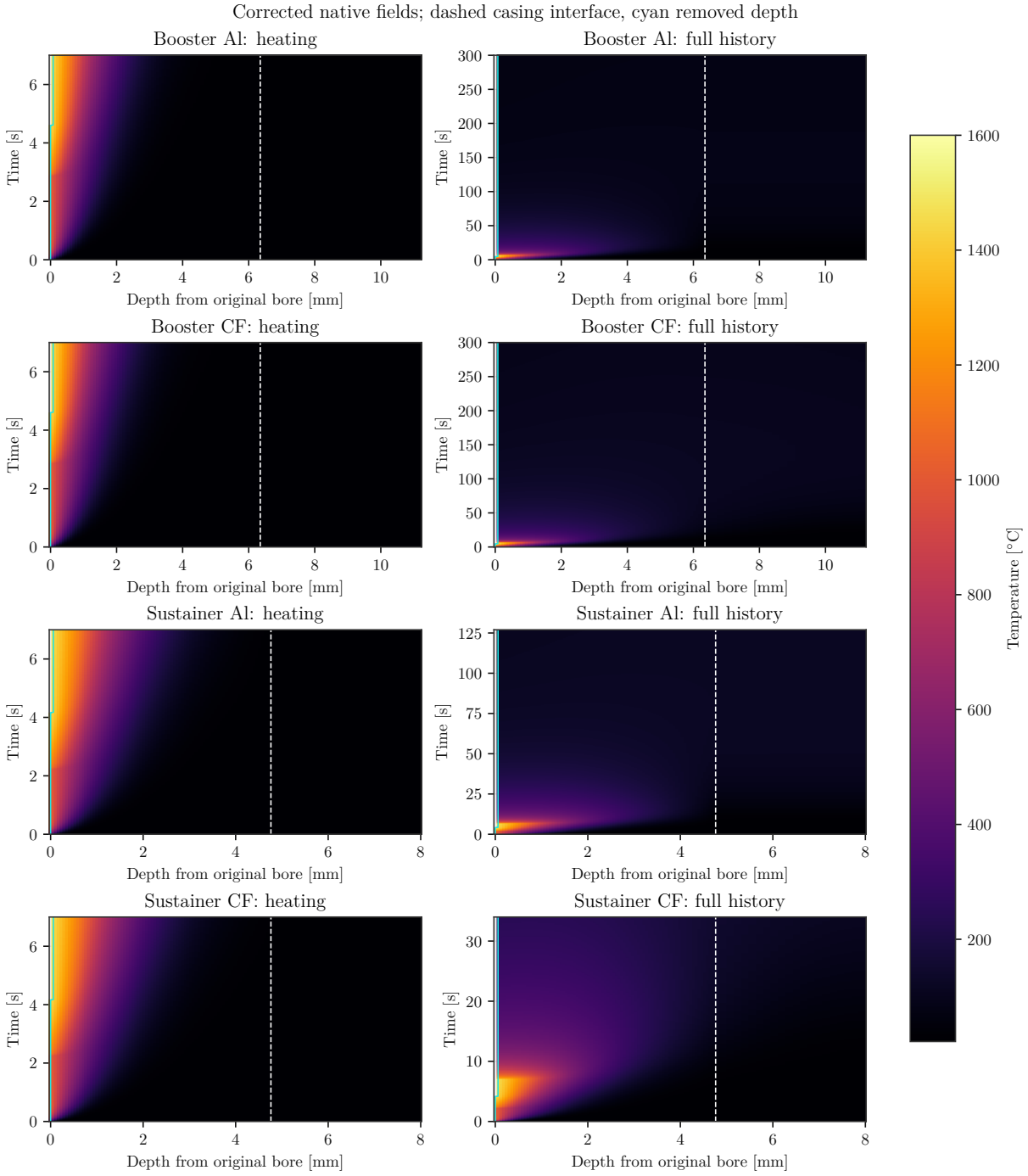


Figure 9. Corrected native temperature fields with a common scale. No time beyond an accepted checkpoint is shown.

10.4 Native temperature fields over the baseline exposure

These historical maps use every accepted checkpoint of the four original twenty-row native baselines, which used the inherited reaction-temperature expression. They are not predictions of the corrected midpoint model. Temperatures

are averaged over each radial station of the uniform sector and interpolated for display. The dashed line marks the liner–casing interface. The cyan line is the upper-cell edge of the contiguous 98-percent conversion front; it is neither recession nor burnthrough. All four coarse baselines retain every liner row through seven seconds, despite nonzero gas and char losses.

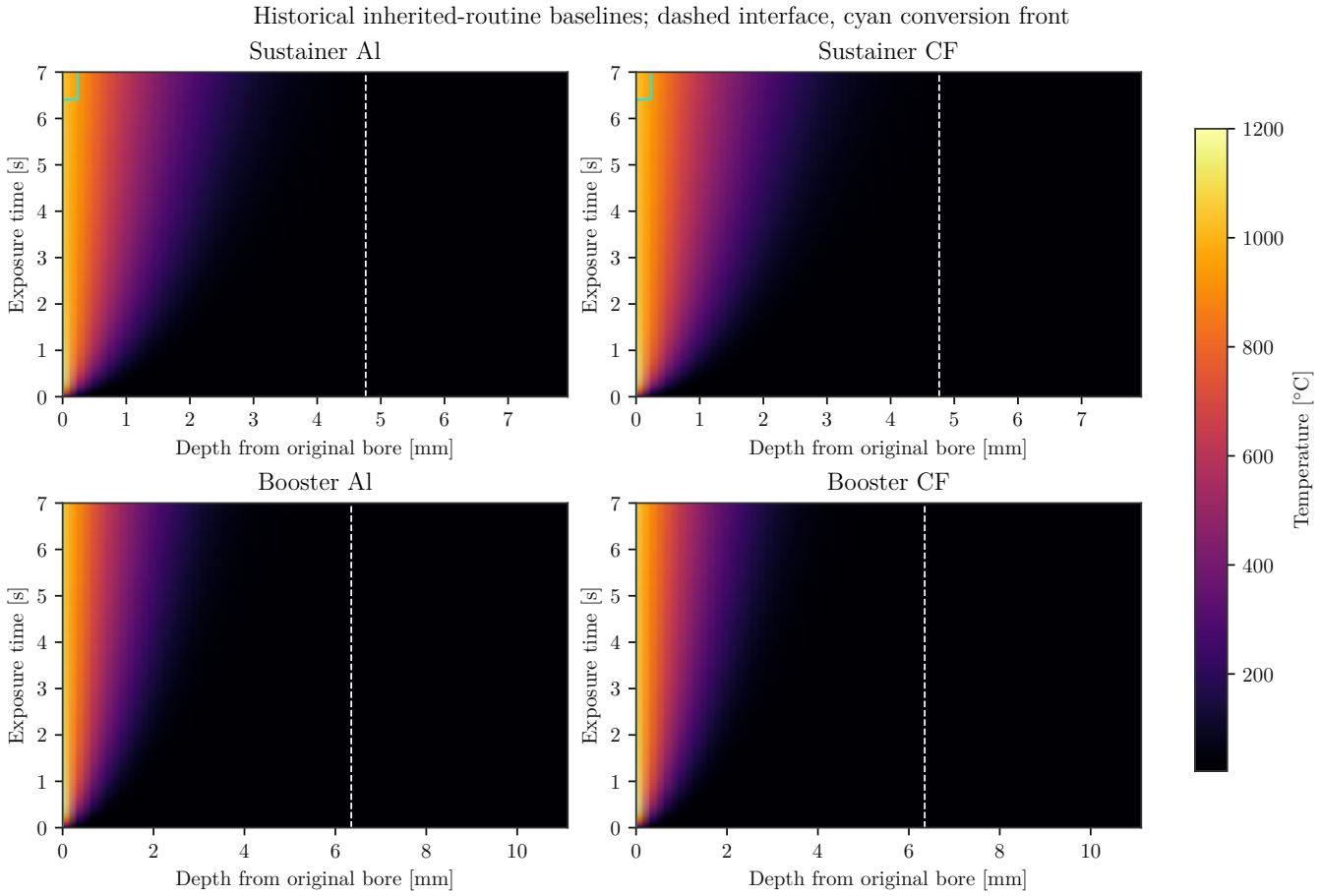


Figure 10. Native ANSYS baseline temperature fields, with a common temperature scale. These coarse-mesh fields are accepted numerical states; the recession discretization still requires the refined comparisons.

11 Independent numerical checks and scope limits

The following historical checks compare a separate cylindrical finite-element assembly with the original coarse ANSYS profiles at 0.1, 1, 2, 5 and 7 seconds. These native baselines used the inherited reaction-temperature expression. Passing the coarse-mesh comparison did not expose that defect; these records remain historical verification evidence, not corrected-model predictions. Numerical agreement cannot establish physical accuracy.

Case	Maximum profile difference [K]	Maximum conversion difference
Sustainer Al	1.0159	0.00263
Sustainer CF	1.0159	0.00263
Booster Al	1.0346	0.00246
Booster CF	1.0346	0.00246

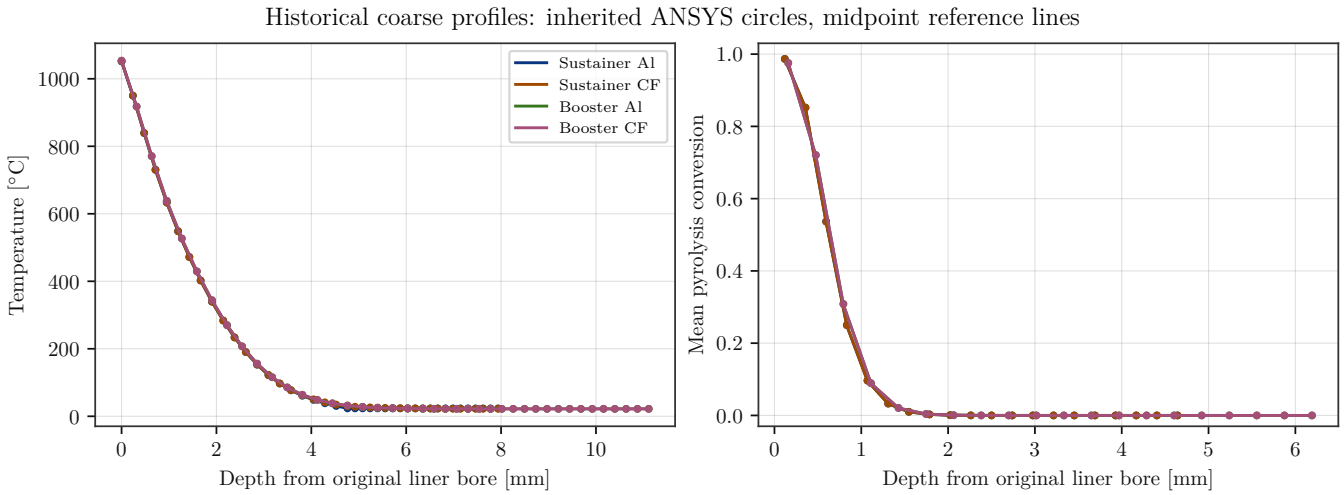


Figure 11. Accepted coarse-mesh native profiles and independent calculations. Temperatures are averaged over each uniform radial station. Agreement on the same mesh is not a mesh-convergence test.

11.1 Refined continuous-exposure stopping points

Under the nominal constant-heating assumptions, the independent model reaches the provisional casing property-range flag while a substantial liner remains. The table uses 320 cells per material and a 0.00078125 s step ceiling. These are model-scope stopping times, not burnthrough times, material failure temperatures or approved operating durations.

Case	Property flag [°C]	Scope ends near [s]	Retained mesh [mm]
Sustainer Al	300	49.485	3.0361
Sustainer CF	150	18.344	4.4053
Booster Al	300	87.274	2.9170
Booster CF	150	32.708	5.4570

The 160-to-320-cell joint refinement changes these stopping times by less than one percent. Separate 80-cell time-step comparisons change them by at most 3.6 percent. Remaining-thickness differences between the finest spatial cases are at most 0.020 mm. These numerical checks concern the property-scope event; they do not validate a later exhaustion event.

11.2 Native layer-removal verification

Sixteen manufactured steady states cover all four configurations with zero rows removed, the first row removed, only the final phenolic row remaining, and the entire phenolic removed. Exact live-element counts, moving hot-face powers and analytical cylinder temperatures passed; the largest temperature discrepancy was 0.00047 K. The final stage applies heating to the exposed casing. These are controlled software checks, not a calculated motor recession history.

The first static fixture omitted the required full Newton option and was rejected by MAPDL. The corrected fixture passed; production transient inputs already contained that option. Static load-factor warnings and unused zero outputs for dead-element heat-table entries are retained in the native log. A separate controller fixture verified one-time residual-mass ejection and termination at the final row, without indexing a nonexistent next liner row.

11.3 Input sensitivities within the provisional material scope

Each bar changes one input at 80 radial cells per material and a 0.003125 s step ceiling. The reference is the same-resolution nominal independent calculation. This is a sensitivity screen, not a probability distribution or a combined worst case. The twenty-percent phenolic perturbations and factor-two removal-cost range are engineering choices, not measured confidence bounds.

Pressure changes affect only the assumed char-reaction closure here; imposed gas temperature and heat-transfer coefficient remain fixed. A small pressure effect in this model does not imply that real motor pressure is unimportant. The NASA alternative uses its slide-14 carbon-epoxy thermal-model tuple: density 1596, conductivity 0.3832 and specific heat 886 in SI units, with a resistance model based on 39 percent resin and 61 percent fibre by volume [8]. It is a different composite, not an identification of the Iris casing.

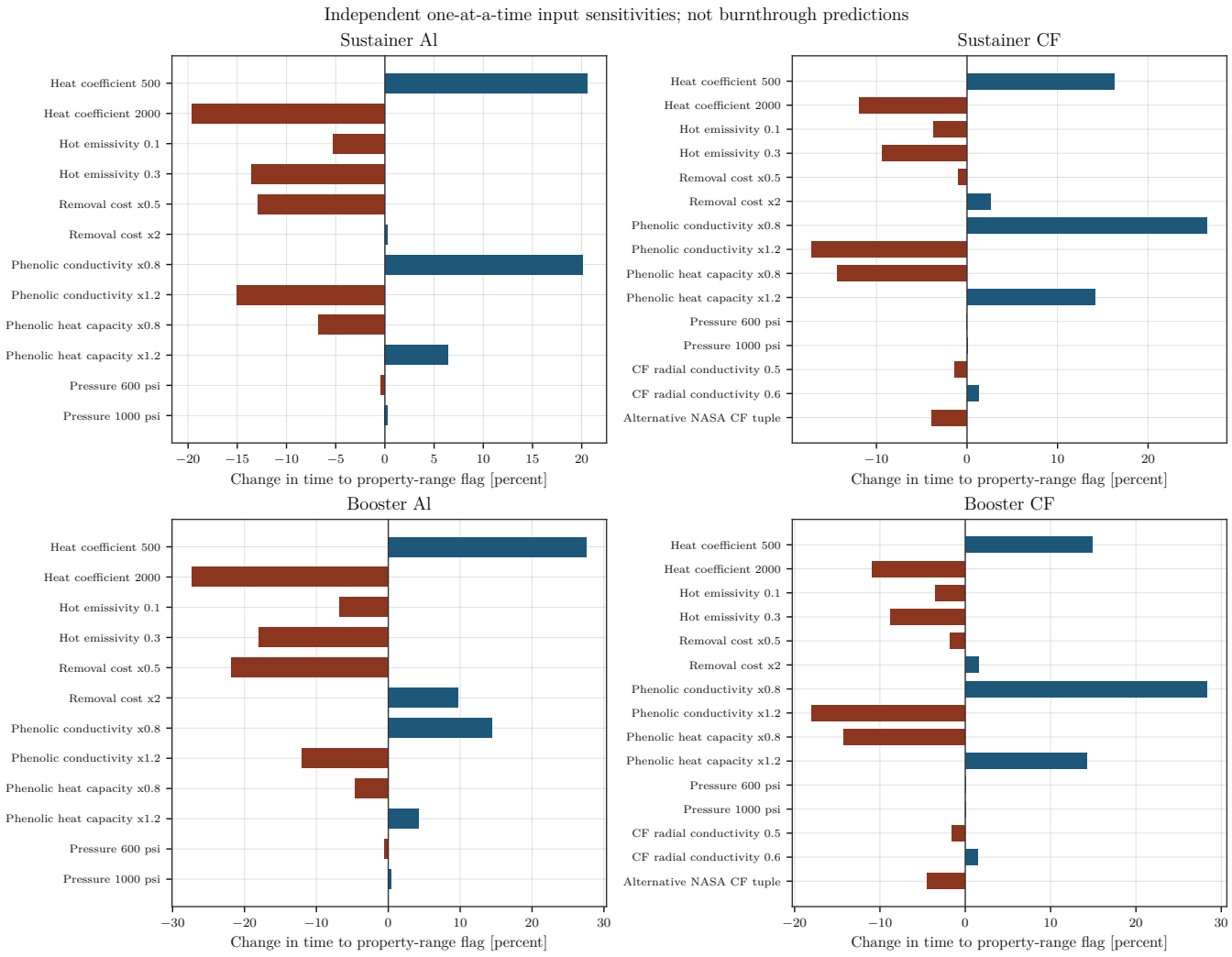


Figure 12. Changes in the duration of prescribed constant heating before the provisional property-range flag. Red bars reach the flag sooner, blue bars later. These bars are not safe firing durations or burnthrough margins.

11.4 Numerical resolution of the sensitivity screen

Ten stronger-heating, lower-phenolic-conductivity and alternative-CF scenarios were repeated with 160 radial cells and half the step ceiling. Their property-scope stopping times changed by at most 1.39 percent. This checks selected scenarios, not every bar. Changes comparable to percent-level discretization error are not resolved physical differences.

Remaining thickness was checked separately. The initial 80-to-160-cell comparisons exceeded the stated thickness tolerance in three scenarios, despite agreement in stopping time. Those cases received another refinement rather than being declared converged from timing alone.

Case	Perturbation	Timing change [%]	Thickness change [mm]	Target [mm]
Sustainer Al	Higher heating	3.50	0.3870	0.0476
Booster CF	Higher heating	0.72	0.0794	0.0635
Booster CF	Lower conductivity	0.48	0.0000	0.0635

This table compares 160 with 320 cells, with a further halving of the time step. The remaining-thickness failures in this comparison prompted the smaller-step investigation below.

11.5 Resolution of the stronger-heating coupling error

The two unresolved doubled-heating cases were recomputed at 320 cells with time steps of 0.0001953125 and 0.00009765625 s, and at 640 cells with 0.000048828125 s. The joint comparison scales the time step with radial

pitch squared. All reported time-only and joint comparisons now meet the five-percent timing target and the retained-thickness target of the larger of 0.02 mm and one percent of initial liner thickness.

Case	640-cell scope stop [s]	Retained [mm]	Joint timing change [%]	Thickness change [mm]
Sustainer Al	39.8790	1.9422	0.100	0.0223
Booster CF	29.7066	4.4648	0.102	0.0198

These finer independent values supersede the coarse stronger-heating recession estimates. They resolve the tested numerical comparisons, not the unknown char kinetics or actual heat load. A closed mass ledger alone did not reveal the earlier time-coupling error. The retained FE thickness remains a whole-row geometric quantity, not a measured intact liner thickness.

11.6 Retained-char heat-capacity diagnostic

Twelve paired independent checks compare the existing full-cell capacity with an optional capacity-only reduction for consumed surface char, at 80, 160 and 320 phenolic cells. Each pair has the same mesh, time schedule and seven-second pulse. The variant changes heat capacity only: conductivity, geometry and source laws remain fixed. It is not a conservative enthalpy-removal model, a validated alternative, or a rigorous uncertainty bound.

Case (320 cells)	Peak change [K]	Scope-time change [%]	Thickness change [mm]
Sustainer Al	-0.265	No flag	0.0000
Sustainer CF	Censored	0.530	0.0000
Booster Al	-0.348	No flag	0.0000
Booster CF	-0.453	No flag	0.0000

At 320 cells the complete-history inner casing peaks change by less than 0.46 K, while the sustainer CF scope time changes by 0.53 percent; no paired retained-thickness change occurs. The maximum consumed char still represented in a live FE row is about 0.143 percent of initial liner mass. At 80 cells, the sustainer CF flag shift is 3.74 percent, so this approximation is more visible on the coarser mesh. These conditional numerical effects do not bound unmodelled cracks, spallation or actual material behavior.

11.7 Native prescribed-temperature reaction check

The native user material was held at uniform prescribed phenolic temperatures of 300, 600 and 1000 degrees Celsius. Conversion at 0.1, 1 and 5 s matched the analytical first-order Arrhenius solution within 1.7e-08 absolute conversion. This verifies the implementation under controlled temperature; it does not validate reaction constants against the actual liner. No surface-consumption controller was used in this fixture.

11.8 Rejected refined step and restart correction

The first 80-row sustainer Al run stopped at an attempted 4.175 s: its energy residual was 10.420 percent, above the unchanged five-percent limit. Its last accepted state was 4.1625 s. There was no internal substep cutback. Investigation found that an earlier removed layer returned after a multiframe restart because the controller reissued only newly pending deactivations. The rejected state is excluded from all accepted results.

ANSYS documents that multiframe restart files do not save element birth/death commands; these must be reissued after restart. The corrected controller replays the entire removed prefix, checks the native live-element count before every solve, and stores two hash-verified accepted native restart generations. See the ANSYS 2026 R1 Basic Analysis Guide, multiframe restart limitations.

The failed runtime and original queued packages are preserved. Fresh sibling packages use identical physical inputs and audit tolerances. The old native restart ring lacked the recent accepted state, so a failed native state was not combined with an earlier controller checkpoint.

A manufactured transient regression reproduced the old resurrection defect for all four cases. Full deactivation replay retained the correct live-element counts and exactly matched uninterrupted nodal temperatures at all sixteen checkpoints. The uninterrupted reference remains in the solution processor until its load steps finish; two earlier fixture drafts incorrectly re-entered a new solution and are preserved as rejected diagnostics. This regression verifies restart topology, not ablation physics.

11.9 Cooling withdrawn after an internal-substep audit gap

The corrected restart run completed seven seconds of heating with one native substep per controller interval. At the first cooling interval, ANSYS cut back the requested 0.25 s step and solved at 7.125, approximately 7.19483, and 7.25 s. The old controller combined the final internal-substep capacity and gas-rate exports with the conversion increment over the whole controller interval. Passing its numerical threshold did not establish a consistent interval energy balance.

The entire subsequent cooling chain is therefore withdrawn from accepted results, and the original run was paused and preserved. The heating prefix through 7 s is retained after inspection of every native load-step entry. Only the final result of each controller interval was saved, so the missing intermediate cooling energy terms cannot be reconstructed faithfully after the run.

Fresh packages export native load-step and substep identity. Acceptance now requires exactly one native substep for each controller interval. The first second of cooling uses 0.0625 s intervals before returning to 0.25 s; any further internal cutback stops acceptance and preserves the preceding accepted restart. Separate default and tighter heat-flow convergence controls use identical schedules through 20 s. Their completed comparison is reported below; the acceptance gates were not relaxed.

A native read-only result probe returned substep 1 at 7 s and substep 3 at 7.25 s; the new acceptance function admits the former and rejects the latter. Retrospective log inspection found only single-substep intervals in all four published coarse baselines and the earlier restart/pulse controls. Their published acceptance is unaffected.

11.10 Reaction-temperature convention and nonlinear tolerance

The intended temporal midpoint is $(T_{n+1}+T_n)/2$. The inherited routine instead added half the temperature increment to the ANSYS temperature argument. The ANSYS UserMatTh interface supplies the current temperature and its increment. Accepted native-temperature replay identifies the actual evaluation as $T_{n+1} + \Delta T/2$, rather than the intended $T_{n+1} - \Delta T/2$. This affects both conversion and the gas sensible-energy sink.

Across the completed 20 s control, independent integration driven by every saved native temperature reproduces conversion with maximum absolute error 2.89e-08 for the inherited expression. The intended midpoint expression differs by up to 0.01819. This isolates the interface convention from differences in the temperature solver.

The earlier isothermal reaction checks could not detect this error: the temperature increment was zero. Their successful constant-temperature results remain valid. The historical native curves elsewhere in this report retain their actual inherited routine and source hashes; they must not be interpreted as verification of the intended midpoint implementation. Earlier independent refinement and sensitivity calculations already use the intended midpoint.

Six additional native prescribed-temperature checks include increasing, decreasing and constant temperatures, each over one 0.01 s step. The inherited library matches the shifted expression; the separately rebuilt candidate matches the intended midpoint. Neither check calibrates reaction constants against a real liner.

Start [°C]	End [°C]	Intended conversion	Inherited native	Corrected native
300	600	7.83092e-05	0.00178077	7.83092e-05
600	1000	0.00252853	0.0176384	0.00252853
1000	600	0.00252853	3.54829e-05	0.00252853
600	300	7.83092e-05	4.09532e-08	7.83092e-05
600	600	0.000488519	0.000488519	0.000488519
1000	1000	0.00779275	0.00779275	0.00779275

The correction changes only the sign of half the increment in the material routine. A new Intel Fortran x64 library was built in an isolated study folder; archived failed and historical runtimes are unchanged. Fresh packages must pass native audits and independent comparison before their thermal results supersede the inherited histories.

The corrected 80/20-cell sustainer Al motor pilot completed eight audited intervals through 0.1 s. Its largest native/independent temperature-profile difference was 0.00124 K. This permits longer corrected verification runs; it does not establish convergence of recession or a complete pulse/cooling history.

Both separate default/tighter nonlinear-tolerance controls completed 20 s with 624 accepted single-substep intervals each. Persistent layer removal, cooling boundary transfer and native balance checks passed. Their maximum live-temperature difference was 6.986 K. Tightening the heat-flow tolerance is a separate numerical correction; it cannot repair the reaction-temperature expression.

Native tolerance	Versus intended midpoint [K]	Versus inherited expression [K]
Default	8.638	6.757
Tighter	5.173	0.229

The last column repeats the independent assembly with the inherited shifted temperature solely to diagnose the old native implementation. It does not replace the intended midpoint model, and agreement is not physical validation. Exact row-event scheduling remains a separate check.

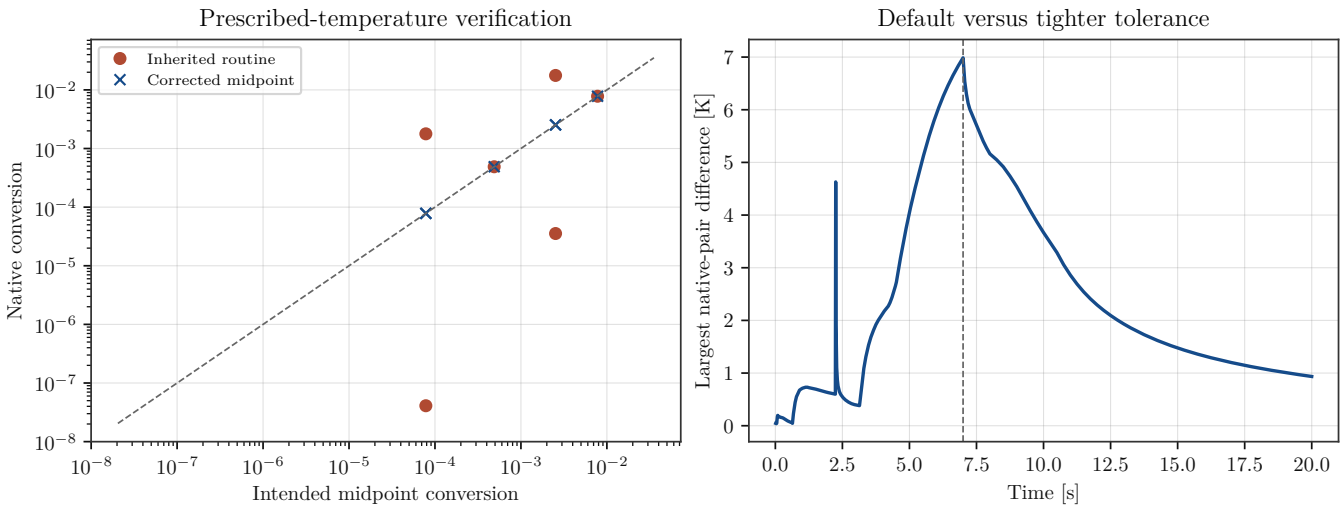


Figure 13. Separate software checks of the reaction-time convention and nonlinear solver tolerance. The dashed diagonal is exact midpoint agreement; the vertical line marks the imposed seven-second heating pulse. These are numerical verification results, not measured motor behaviour.

11.11 Refined native pulse cross-checks

These comparisons use the independent assembly at the same 80/20 radial mesh and coupling schedule as the native pulse cases. Only accepted native checkpoint times are compared; a running case has no predicted ANSYS future values. Layer-removal scheduling and the next hot-face transfer are checked separately.

Case	Through [s]	Max profile [K]	Max conversion	Transfers passed
Sustainer Al	127.000	0.333	0.00027	1
Sustainer CF	34.000	0.333	0.00027	1
Booster Al	300.000	0.220	0.00016	1
Booster CF	300.000	0.220	0.00016	1

All displayed comparisons meet the same-mesh profile, conversion, mass and row-count checks.

These checks do not establish fine-mesh convergence of the native stopping time. In particular, the independent sustainer CF pulse needed further refinement of its provisional property-flag time.

This table uses fresh corrected-midpoint native runs. The earlier seven-second historical comparison and withdrawn cooling chain are excluded. Agreement now checks the intended midpoint reaction model, persistent layer deactivation and the strict single-substep cooling transition together.

Case	Row	Native scheduled [s]	Independent [s]	Difference [s]
Sustainer Al	1	4.1625	4.1625	0.00000
Sustainer CF	1	4.1625	4.1625	0.00000
Booster Al	1	4.6000	4.6000	0.00000
Booster CF	1	4.6000	4.6000	0.00000

These are individual row-removal events, not liner-exhaustion times. The native event is bracketed by its previous accepted intact state and the scheduled state; geometric deactivation occurs in the next solve.

11.12 Variation within the uniform sector

Every accepted checkpoint in the completed corrected primary cases was checked for differences among the nine nodes at each radial station. The table reports the largest axial/angular spread at a common radius, using only nodes belonging to elements active during that solve. This quantifies information suppressed by radial averaging; it does not establish uniformity in a real motor or bound unmodelled end effects.

Case	Through [s]	Phenolic range [K]	Casing range [K]
Sustainer Al	127.00	1.19e-08	1.13e-12
Sustainer CF	34.00	0.00653	0.013
Booster Al	300.00	1.07e-08	4.55e-13
Booster CF	300.00	0.00575	0.0104

The completed 160/40-row corrected-geometry cases were also checked at every accepted checkpoint. Their maximum same-radius ranges are listed separately below; these small ranges do not establish physical uniformity outside the modelled slice.

Refined case	Through [s]	Phenolic range [K]	Casing range [K]
Sustainer Al	127.00	1.36e-08	1.05e-12
Sustainer CF	31.25	1.4e-05	0.000139
Booster Al	300.00	1.2e-08	6.11e-13

11.13 Native spatial and time-resolution program

The original native resolution plan has thirteen additional cases. For each motor/casing pair, the first comparison keeps 80 phenolic and 20 casing rows while reducing the heating and cooling coupling steps by four. The 160/40-row case retains those refined time steps. Its current transition from the older 80-row case also changes sector geometry and solver tolerance, so that pair is labelled as a combined numerical comparison. The corrected 320/80-row case retains the new geometry measures and solver tolerance while halving the time steps. Sustainer CF additionally uses 640/160 rows because the independent 160-to-320-cell scope-time comparison failed its target.

The 80-row time check and 160-row spatial check use 0.003125 s heating, 0.015625 s during the first cooling second, and 0.0625 s thereafter. The 320-row case halves these intervals; the 640-row case halves them again. All use the same physical inputs, corrected library, tighter nonlinear tolerance and unchanged audit gates. Each package preserves the primary cooling horizon and stops at the provisional material flag.

Native comparisons require casing peak changes below 5 K for completed histories, scope-time changes below five percent for censored histories, and retained-thickness changes no larger than the greater of 0.02 mm or one percent of the initial liner. A scope-limited calculation cannot supply a later casing peak. Preparation is not an accepted result.

Case	Peak change [K]	Scope change [%]	Thickness [mm]	Comparison	Pass
Sustainer Al	0.721	–	0.0000	time quarter	Yes
Sustainer Al	2.469	–	0.0000	Combined numerical	Yes
Sustainer CF	–	2.941	0.0000	time quarter	Yes
Sustainer CF	–	10.714	0.0000	Combined numerical	No
Booster Al	0.224	–	0.0000	time quarter	Yes
Booster Al	2.711	–	0.0397	Combined numerical	Yes
Booster CF	0.318	–	0.0000	time quarter	Yes

Only completed, audited native cases with passing same-mesh independent checks enter this table. Earlier refinement failures remain visible; convergence is judged on the final relevant comparisons.

The combined numerical comparisons for Sustainer Al, Sustainer CF, Booster Al change casing radial resolution, sector angle, area/volume formulas, nonlinear heat-flow tolerance, phenolic radial resolution. Their outcomes cannot be attributed to radial mesh refinement alone. The finer comparisons must retain common sector measures and solver controls to isolate the remaining mesh/time effects. Execution-core changes are recorded separately.

Case	Inner peak [°C]	Time [s]	Outer peak [°C]	Time [s]	Retained [mm]
Sustainer Al	119.777	79.1875	119.757	79.2500	4.702969
Booster Al	84.560	142.6875	84.542	142.8750	6.270625
Booster CF	113.925	102.8750	104.866	144.1875	6.270625

Table 6. Accepted native quarter-step casing peaks for histories completing their prescribed cooling horizon, at unchanged 80/20-row spatial resolution. The scope-limited sustainer CF result is described separately below. These values do not establish spatial convergence or a motor lifetime.

Case	Rows	Inner peak [°C]	Time [s]	Outer peak [°C]	Time [s]	Retained [mm]
Sustainer Al	160/40	122.245	78.1250	122.225	78.2500	4.702969
Booster Al	160/40	87.271	141.2500	87.252	141.4375	6.310313

Table 7. Completed finer native histories. Rows lists phenolic/casing radial cells. Values are sampled maxima over each completed cooling horizon; scope-limited CF histories are excluded from this peak table. Retained thickness is a whole-row mesh quantity, not a measurement of intact material.

For Sustainer CF, the 80-to-80-row time quarter comparison changes the accepted material-scope bracket from [33.7500, 34.0000] s to [34.9375, 35.0000] s. The refined endpoint has an inner casing temperature of 150.013

degrees Celsius and retains 4.702969 mm of mesh. These are cooling-phase property-range stops, not burnthrough or structural failure. Their outer casing endpoint temperatures are not completed cooling peaks.

For Sustainer CF, the 80-to-160-row combined numerical comparison changes the accepted material-scope bracket from [34.9375, 35.0000] s to [31.1875, 31.2500] s. The refined endpoint has an inner casing temperature of 150.058 degrees Celsius and retains 4.702969 mm of mesh. These are cooling-phase property-range stops, not burnthrough or structural failure. Their outer casing endpoint temperatures are not completed cooling peaks.

The scope-time change is 10.714 percent, which exceeds the five-percent target despite unchanged retained mesh. This is an accepted numerical solution with an unresolved refinement difference; the finer queued comparisons are still required.

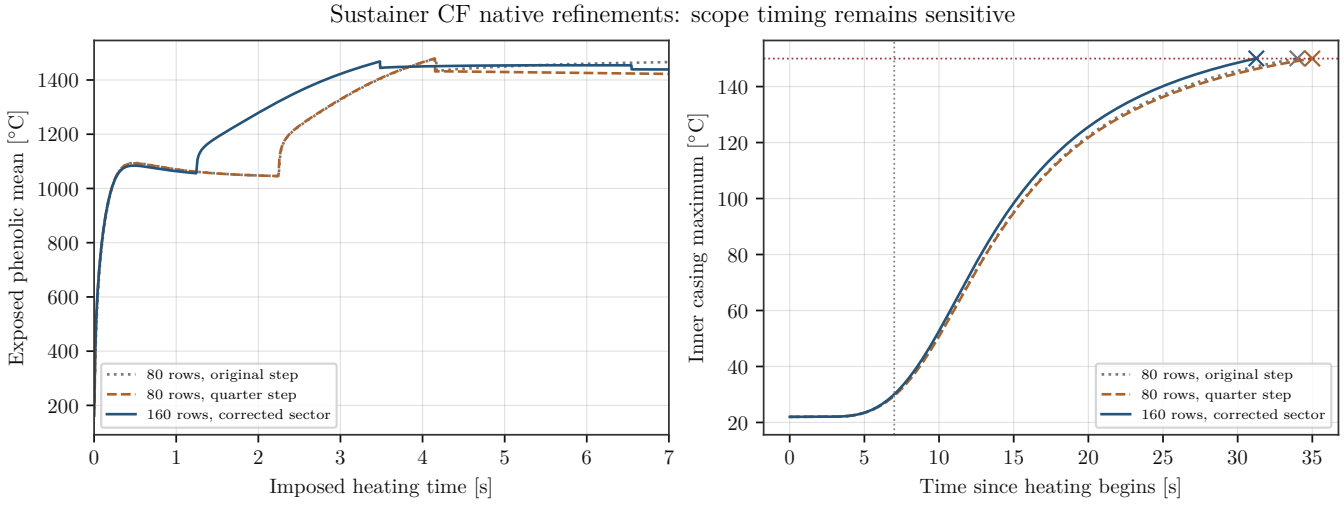


Figure 14. Accepted native sustainer CF histories. Lines join accepted samples; x marks each provisional 150-degree casing-property stop. Left: exposed phenolic during the seven-second heating pulse. Right: delayed inner-casing heating. The newer 160-row case also changes sector geometry/measures and solver tolerance, so its difference is not an isolated radial-mesh error. No curve is extended past its scope stop.

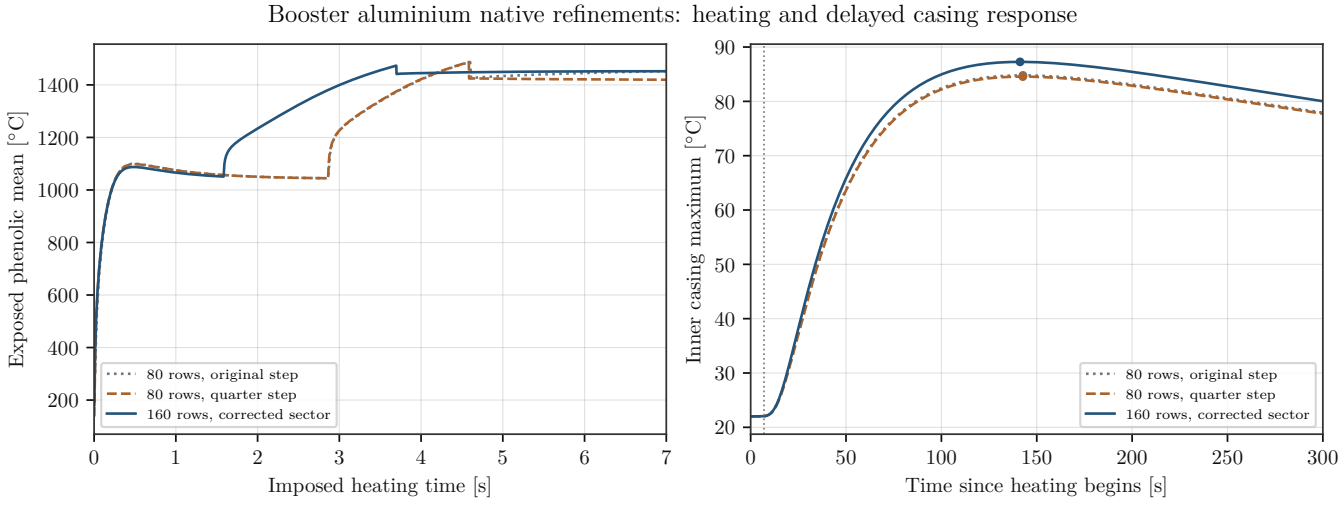


Figure 15. Completed native booster aluminium histories. Lines join accepted samples; dots mark sampled inner-casing peaks. Left: exposed phenolic during the imposed seven-second pulse. Right: casing heating and cooling through 300 s. The 160-row calculation also changes sector geometry/measures and solver tolerance. Passing casing and thickness targets does not establish hot-surface or isolated radial-mesh convergence.

6 prepared native resolution cases remain without terminal accepted comparisons at this edition. The primary histories remain separate from these refinements; the hot-face time comparison is reported earlier.

11.14 Sensitivity to the row-removal cutoff

The removal threshold is a numerical choice: a row is deactivated when residual solid mass is less than 0.1 percent of its initial virgin mass. Twenty-four paired independent runs tested 0.01, 0.1 and 1 percent cutoffs for all four cases at 160/40 and 320/80 phenolic/casing cells. Physical laws and time settings were fixed within each pair. These are numerical-tolerance perturbations, not material uncertainty.

Case (320 cells)	Cutoff [%]	Inner peak change [K]	Scope-time change [%]	Retained change [mm]
Sustainer Al	0.01	2.399	–	0.0149
Sustainer Al	1	-3.812	–	-0.0298
Sustainer CF	0.01	Censored	-7.211	0.0149
Sustainer CF	1	Censored	16.119	-0.0298
Booster Al	0.01	0.973	–	0.0000
Booster Al	1	-2.636	–	-0.0397
Booster CF	0.01	1.406	–	0.0000
Booster CF	1	-3.716	–	-0.0397

Changes are relative to the 0.1-percent calculation on the same mesh. At 320 cells, complete-history casing peaks shift by as much as 3.812 K. Sustainer CF scope timing changes by about -7.21 to +16.12 percent, exceeding the five-percent timing target. This sensitivity is separate from radial/time convergence at a fixed cutoff. The queued native cases retain their sealed 0.1-percent rule; their reported times remain conditional on that choice.

The small expelled residual mass does not make this timing effect negligible. A nearly consumed row can remain thermally active while the residual virgin fraction continues reacting, delaying transfer of the hot boundary to the next row. This explains why a mass-ledger check alone does not settle removal discretization. No real-motor lifetime follows from these cutoff-dependent states.

The chosen equations also explain why a zero cutoff is not a physically validated target. Char consumption cannot exceed the available char, so $m_{\mathrm{solid}}/V \geq \rho_v(1-\alpha)$. The first-order conversion law leaves $1-\alpha = (1-\alpha_0)\exp(-\int k\,dt) > 0$ at finite integrated reaction exposure. Thus exact zero residual is not reached in finite time in the continuous equations. The finite cutoff approximates ejection or loss of continuity of the residual solid, for which this study has no calibration. Tighter cutoffs test numerical/model dependence; they do not automatically improve physical accuracy.

Additional fixed-640-cell sustainer CF checks at fractional cutoffs 1e-7 and 1e-8 change the scope time by 1.369 and 1.514 percent relative to the preceding decade. The 1e-8 case stops at 24.3906 s with 4.7551 mm retained. These smaller successive changes do not erase the difference from the original cutoff or identify a physically correct residual-ejection rule.

Additional independent calculations now compare successive fractional cutoffs of 1e-4, 1e-5 and 1e-6 at 320 and 640 cells. These tests concern the numerical equations, not physical validation of the removal law.

Case	Final comparison	Peak change [K]	Scope change [%]	Retained change [mm]	Pass
Sustainer CF	Cutoff	–	-2.606	0.0074	Yes
Sustainer CF	Mesh/time	–	-2.488	0.0000	Yes
Sustainer Al	Cutoff	1.078	–	0.0074	Yes
Sustainer Al	Mesh/time	0.868	–	0.0000	Yes
Booster Al	Cutoff	0.490	–	0.0000	Yes
Booster Al	Mesh/time	0.522	–	0.0000	Yes
Booster CF	Cutoff	0.705	–	0.0000	Yes
Booster CF	Mesh/time	0.745	–	0.0000	Yes

Passing a successive-refinement target does not prove a zero-cutoff limit or remove the difference from the original 0.1-percent choice. Native confirmation and interpretation of the tighter rule remain separate work; no queued sealed configuration is silently replaced.

A separate native 80/20-row sustainer CF control with fractional cutoff 1e-6 has been prepared but has not completed both verification gates. Its time schedule matches the 80-row quarter-step default-cutoff reference, allowing a controlled later native comparison. Neither sealed input is changed in place.

11.15 Combined-input pulse scenarios

Two joint engineering perturbations were calculated for all four configurations, separately from ANSYS. The first doubles the heat-transfer coefficient to 2000 in SI units, raises phenolic conductivity by 20 percent and lowers its heat capacity by 20 percent. The second doubles the same coefficient, lowers phenolic conductivity by 20 percent and halves the effective char-removal cost to 17.5 MJ/kg. Every change comes from the earlier one-at-a-time engineering screen; their combination is neither a measured uncertainty interval nor a demonstrated worst case.

All cases use a hypothetical seven-second heating pulse with the primary ambient-cooling assumptions and property-scope stops. Eight scenarios were each repeated with 80/20, 160/40 and 320/80 phenolic/casing cells.

Heating steps scale with radial pitch squared: 0.003125, 0.00078125 and 0.0001953125 s. Later cooling steps are respectively 0.0625, 0.015625 and 0.00390625 s. Intermediate time and mesh effects remain coupled in this additional input screen; the separate native resolution program tests those effects independently.

All eight final comparisons meet the stated targets. Complete-history casing maxima change by at most 0.855 K; scope-time changes are at most 0.259 percent. Retained-thickness checks pass separately. These numerical results do not calibrate the joint physical assumptions.

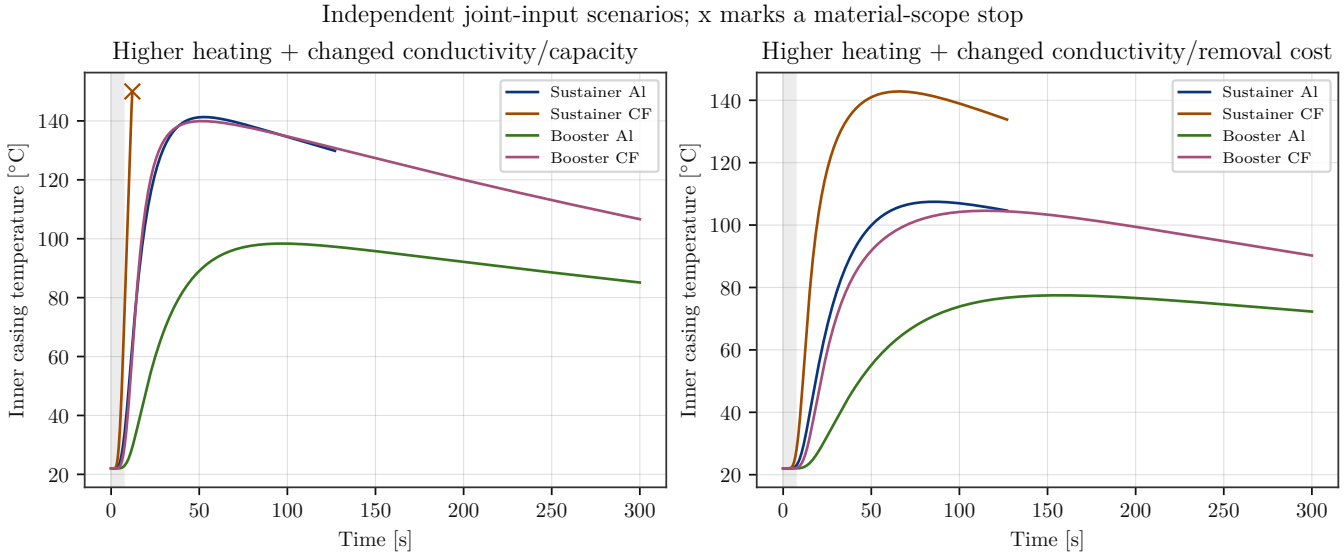


Figure 16. Independent 320-row combined scenarios. The shaded interval is imposed heating. Scope-limited curves do not predict a later casing peak or structural failure.

Case	Joint scenario	Through [s]	Inner max [°C]	Retained [mm]
Sustainer Al	Transport	127.000	141.31	4.5988
Sustainer Al	Consumption	127.000	107.48	4.3607
Sustainer CF	Transport	12.047	150.01	4.5988
Sustainer CF	Consumption	127.000	142.83	4.3607
Booster Al	Transport	300.000	98.36	6.1912
Booster Al	Consumption	300.000	77.50	5.9730
Booster CF	Transport	300.000	139.93	6.1912
Booster CF	Consumption	300.000	104.59	5.9730

Changing several assumptions can change both heat admitted to the wall and energy removed by gas generation and char consumption. A larger imposed heat-transfer coefficient therefore need not produce a hotter casing when the other closure parameters also change. This is sensitivity of the chosen equations, not evidence that changing a real liner would improve safety. No scenario here establishes a real-motor burnthrough time.

Four separate bounded native controls have been prepared for higher heating, the published alternative CF tuple and two combined cases. They stop at 20 s or an earlier material flag and test implementation against the same-mesh independent model. Their short horizons cannot establish later cooling maxima.

At this snapshot, 0 of those four controls have completed both native audit and independent comparison. Prepared or running controls supply no completed native confirmation.

12 Independent pulse and cooling investigation

These curves are independent radial calculations, not ANSYS output. Each uses seven seconds of the prescribed hot-gas boundary followed by ambient gas with a heat-transfer coefficient of 8 in SI units, zero hot radiation and zero char consumption. Pyrolysis and conduction continue during heat soak. The pulse duration and cooling conditions are assumed, not measured firing or tailoff histories.

The sustainer Al and both booster references use 320 phenolic and 80 casing cells. Sustainer CF required 640 and 160 cells to bring its property-flag timing change below five percent. Its curve stops at the provisional 150-degree casing flag, so its later peak is unknown within this material model. The other curves end after their sampled casing peaks.

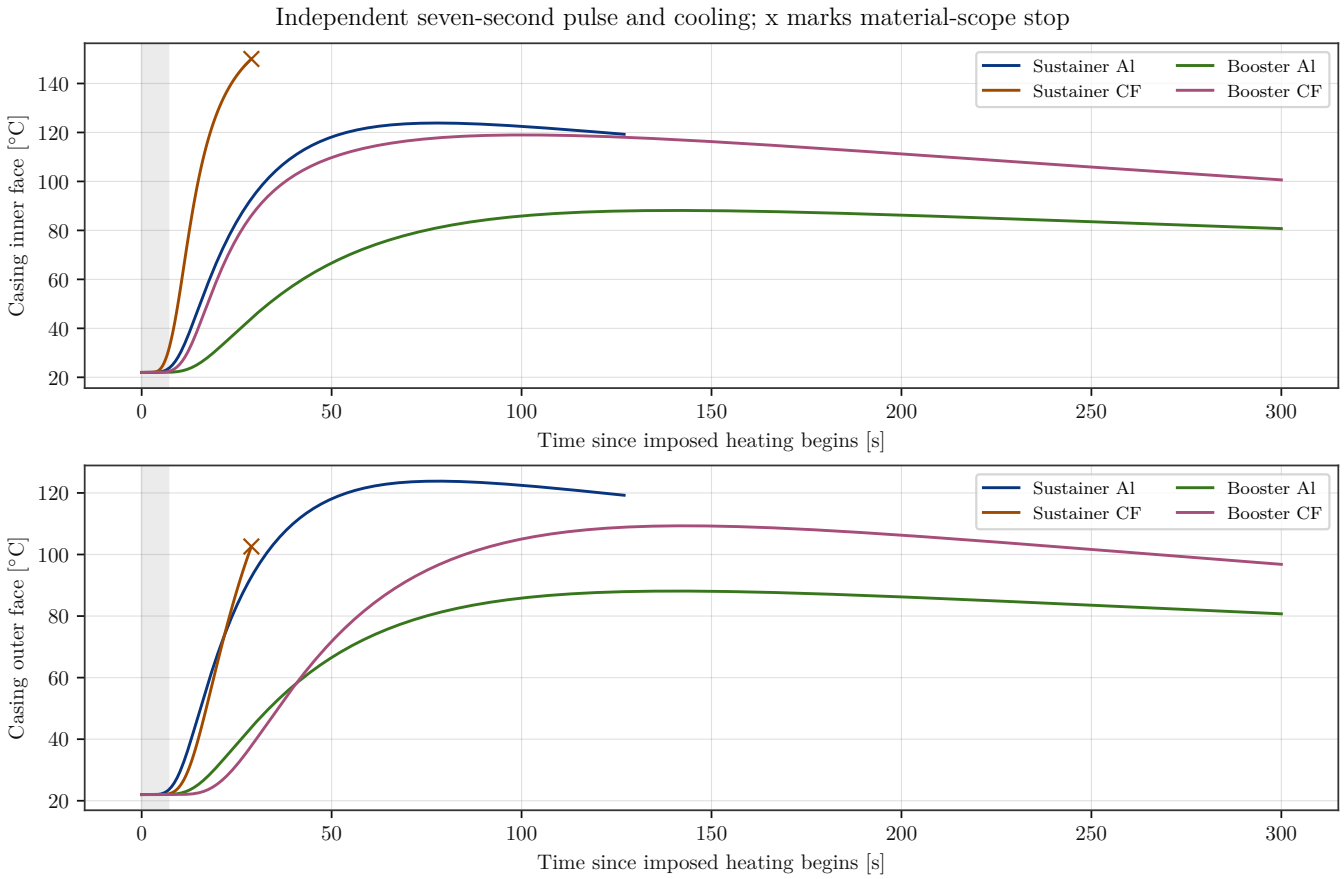


Figure 17. Independent refined temperature histories. The shaded interval is imposed heating; curves beyond it describe assumed cooling. A scope stop is not a predicted casing failure.

Case	Through [s]	Inner max [°C]	Outer max [°C]	Retained mesh [mm]
Sustainer Al	127.00	123.85	123.83	4.7179
Sustainer CF	28.88	150.01	102.60	4.7179
Booster Al	300.00	88.14	88.12	6.3103
Booster CF	300.00	118.99	109.32	6.3103

For the complete 127-second histories, the last 160-to-320-cell joint refinement changes sampled casing maxima by at most 1.7 K and remaining mesh thickness by at most 0.015 mm. Sustainer CF instead compares its censored stopping time; the 320-to-640-cell change is about two percent. These checks do not establish convergence of every hot-face temperature or a future exhaustion event.

Both booster cases still had a rising casing temperature at 127 s. Independent extension to 300 s captured their later peaks, so fresh native booster packages were prepared with that longer cooling horizon. The original unstarted packages are preserved. Native confirmation of the refined pulse/cooling histories remains pending.

12.1 Exploratory twenty- and sixty-second pulses

These durations are assumed scenarios, not firing-time estimates. Each calculation stops at its provisional casing-property flag or continues for 300 s after heating. A stop during heating means the prescribed pulse was not completed within the material model; a stop during cooling identifies delayed heat soak. None reached phenolic exhaustion within the accepted model scope.

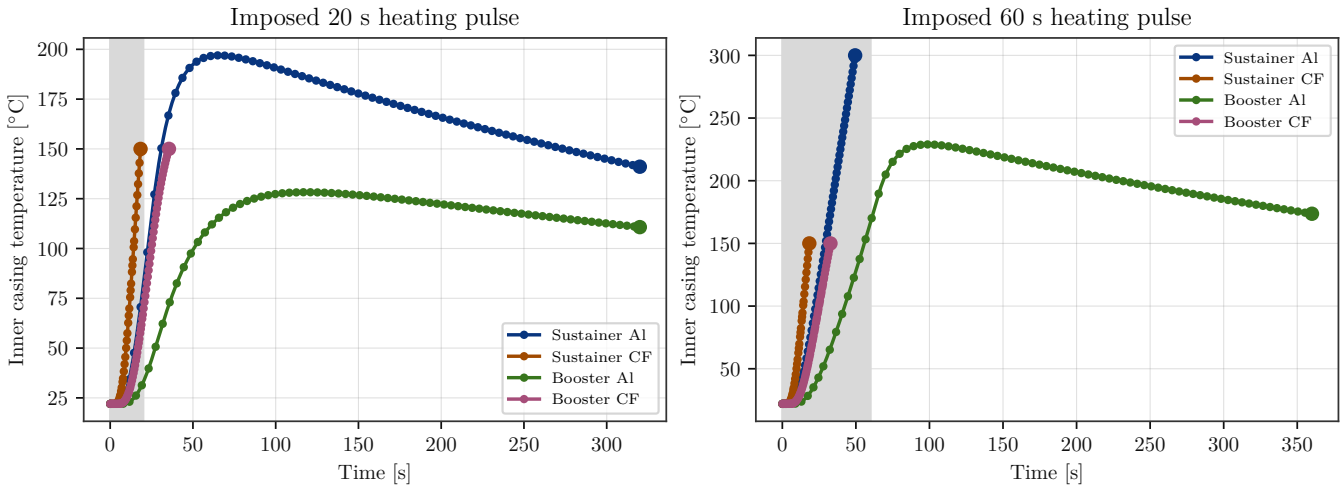


Figure 18. Independent 320-cell longer-pulse scenarios; gray shading marks prescribed heating. Curves end at a scope flag or at the scenario horizon.

Case	Pulse [s]	Through [s]	Stop reason	Inner max [°C]	Retained [mm]
Sustainer Al	20	320.000	Horizon	196.96	4.3458
Sustainer Al	60	49.486	During heating	300.00	3.0361
Sustainer CF	20	18.344	During heating	150.00	4.4053
Sustainer CF	60	18.344	During heating	150.00	4.4053
Booster Al	20	320.000	Horizon	128.25	5.9531
Booster Al	60	360.000	Horizon	228.96	4.1870
Booster CF	20	35.500	During cooling	150.03	5.9531
Booster CF	60	32.709	During heating	150.00	5.4570

All eight 160-to-320-cell comparisons passed their numerical targets. Censored flag times changed by at most 1.05 percent; complete-history casing maxima changed by less than 0.97 K, and retained thickness changed by at most 0.020 mm. Sustainer CF reaches the same scope stop before either longer pulse ends, so those two rows describe the same partial heating history. These independent scenarios still require native confirmation of influential conditions and measured heating/material data for a physical lifetime claim.

13 Geometry consistency and exact removal-event verification

The complete 127-second native geometry control now passes verification. The ten-core sustainer Al calculation completed 4208 accepted intervals, including both removal events and the seven-second pulse followed by 120 seconds of cooling. Terminal logs, sealed inputs, native/controller restart backups, balances and the independent cylindrical comparison pass. This resolves the tested control; the remaining motor/casing refinements are unfinished.

Its sampled inner-casing maximum is 122.2454 degrees Celsius at 78.1250 s; the outer maximum is 122.2249 degrees Celsius at 78.2500 s. Retained mesh thickness is 4.702969 mm. The largest casing-peak shift from the completed 80/20-row quarter-step case is 2.4689 K. That pair changes radial mesh, sector angle, controller measures and nonlinear tolerance together. It is a combined numerical comparison, not an isolated radial-mesh error estimate. The exposed-surface mean at 7 s changes from 1422.376 to 1438.802 degrees Celsius (+16.426 K). Passing casing targets therefore does not establish surface-temperature convergence.

13.1 Why the earlier 160-row comparisons were held

The original 160/40-row mesh uses two angular cells across a 0.5° sector. Its controller assigned circular-arc areas and volumes, while the eight-node SOLID278 mesh has straight-sided faces. For one angular cell of width $\delta\theta$ radians and axial length ℓ, the geometric measures are

$$A_{\mathrm{arc}} = r\,\delta\theta\,\ell, \qquad A_{\mathrm{hex}} = 2r\sin(\delta\theta/2)\,\ell, \tag{25}$$

$$V_{\mathrm{arc}} = \tfrac{1}{2}(r_o^2 - r_i^2)\delta\theta\,\ell, \qquad V_{\mathrm{hex}} = \tfrac{1}{2}(r_o^2 - r_i^2)\sin(\delta\theta)\,\ell. \tag{26}$$

Thus $A_{\mathrm{hex}}/A_{\mathrm{arc}} = \sin(\delta\theta/2)/(\delta\theta/2)$ and $V_{\mathrm{hex}}/V_{\mathrm{arc}} = \sin(\delta\theta)/\delta\theta$. At $\delta\theta = 0.25°$, these ratios are 0.999999206726 and 0.999996826905. Direct triangle-area and hexahedral-Jacobian integration of the actual mesh coordinates confirmed the mismatch. The native applied hot-face power also followed the polygonal area rather than the controller's arc area. These discrepancies are small, but the discrete removal event is triggered by crossing a residual-mass threshold; a small accumulated difference can move it by a complete coupling interval.

The original native calculation scheduled row two at 6.540625 s, whereas the independent cylindrical calculation scheduled it at 6.543750 s: one heating interval, $\Delta t = 0.003125$ s, later. Tightening the native heat-flow tolerance from 10^{-5} to 10^{-7} did not change this mismatch. Both native runs passed their balance and topology audits but failed the strict event comparison, and were cleanly paused with their records preserved. Their selected live-profile differences were below 0.25 K; field agreement alone therefore was insufficient. An isolated independent area perturbation reproduced the earlier second event, supporting the geometry diagnosis without establishing a unique cause for every remaining field difference. No material coefficient or removal cutoff was fitted.

13.2 Fresh control and unchanged acceptance rules

The fresh control retains 160 phenolic and 40 casing radial cells, two axial and two angular cells, 800 elements and 1809 nodes. Height remains 0.5 mm. The total angle is reduced to 0.05° and the controller now uses the exact straight-sided measures above. Full-ring mass/power scaling becomes 7200. The radial and temporal resolution, 10^{-7} heat-flow tolerance, material DLL, reaction laws, 0.001 residual-solid cutoff and comparison gates are unchanged from the tighter-tolerance diagnostic. Initial temperature is 22 degrees Celsius; no discrepant native solution was inherited.

This is a *joint geometry-consistency and angular-refinement control*; the two changes have not been tested separately. Direct integration agrees with its controller areas to 3.3×10^{-13} relative and volumes to 1.1×10^{-10}. A separate two-solid curved STEP passes the dimension and volume round-trip checks; its exact CAD volumes are deliberately distinguished from the straight-sided finite-element volumes. Historical primary maps and sealed inputs were checked and remain unchanged.

13.3 Verified full history, events and field comparisons

The removal gate requires matching event sets and matching scheduled times within 10^{-8} s, in addition to matching removed-row counts. Either sign of a one-interval mismatch fails. A regression check retained all nine previously passing comparisons. For this complete control both event sets match, and the largest event-time difference is 1.56×10^{-13} s (floating-point accumulation). Scheduling occurs after a converged interval; the new exposed face receives the load during the following solve, hence the distinct transfer times below.

Row	Interval	Native schedule [s]	Reference schedule [s]	First transfer [s]
1	1114	3.481250	3.481250	3.484375
2	2094	6.543750	6.543750	6.546875

Table 8. Whole-row removal events in the completed verified control. The second native event lies in the bracket (6.540625, 6.543750] s.

At 7 s the imposed hot-gas pulse ends. The subsequent solves use 22-degree gas and $h = 8\ \mathrm{W\,m^{-2}\,K^{-1}}$, with char consumption disabled by the prescribed exposure switch. Native load transfer, persistent deactivation and cooling controls pass. This switch verifies the imposed scenario, not a measured motor tailoff.

All 4208 native intervals through 127.0000 s pass the step-balance, live-element, single-substep and sealed-input checks. Maximum relative energy residual is 7.994×10^{-5} and hot-power residual is 5.72×10^{-8}; maximum absolute mass residual is 7.737×10^{-22} kg. Immutable audit/checkpoint hashes and the selected independent profiles are frozen in `validation/native_geometry_control_terminal.json`. Its hashes identify the full terminal native audit and independent comparison; the earlier frozen prefix remains preserved separately.

Across 20 selected comparison times, including both schedule/transfer pairs, the pulse switch and the final accepted state, the largest live-profile difference is 0.00358855 K, at 7 s. This is a maximum over the recorded comparisons, not a bound over every intermediate native state. Conversion and solid-mass checks also pass. The reference remains a separate cylindrical radial assembly; numerical agreement does not calibrate the unmeasured wall heating or char law.

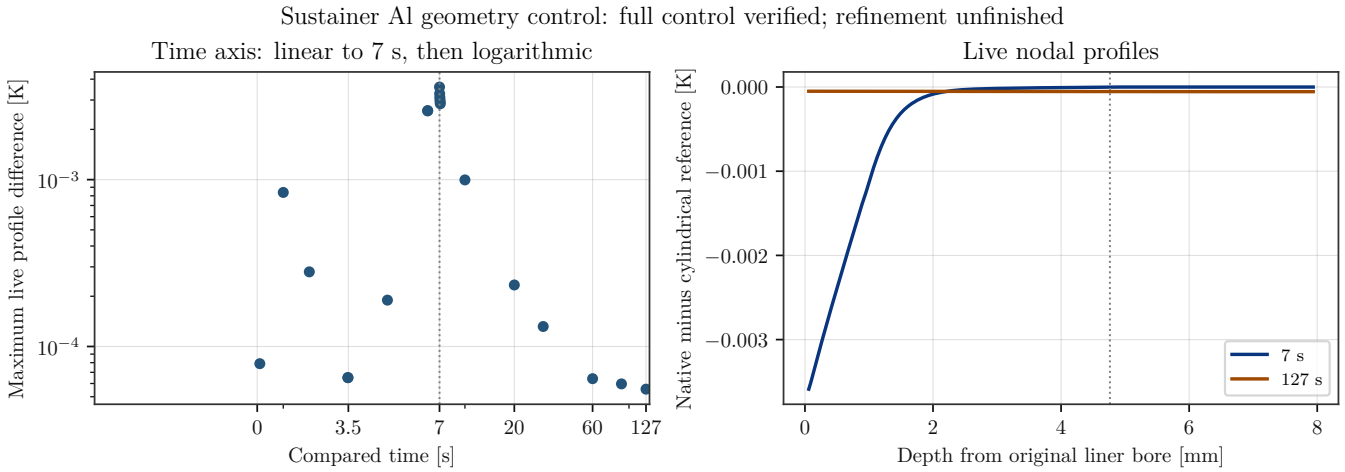


Figure 19. Frozen native/reference comparison. The left panel shows discrete sample comparisons, without interpolation between them; its time axis is linear through 7 s and logarithmic thereafter. The right panel excludes removed rows; its vertical dotted line marks the phenolic/casing interface.

The verified control permits the remaining native refinement cases to use fresh geometry-consistent packages with the same sector measures and nonlinear tolerance. 6 further resolution cases and five later controls remain. Only one ten-core case runs at a time; superseded packages remain preserved. The primary and time-only result sets are unchanged. Neither this numerical agreement nor the retained mesh establishes a calibrated motor lifetime.

14 Next results

Complete the corrected native refined pulse/cooling cases and the remaining native resolution and influential-input checks. Compare persistent first-row removal, cooling, temperature profiles and mass loss against the independent assembly, within the stated material scope. Report whole-cell recession separately from the 98 percent pyrolysis front. Preserve rejected diagnostics with their reasons rather than silently treating them as successful predictions.

A native one-second heating pulse followed by five cooling steps passed the boundary-transfer and balance checks. This verifies the prescribed switch to ambient gas and zero char consumption; it does not measure real tailoff.

A defensible final answer may be a conditional “not reached through the accepted exposure” result with remaining thickness, rather than a finite lifetime. Neither missing wall-exposure data nor uncalibrated char kinetics can be recovered from solver convergence. The study cannot establish real motor lifetime without supported wall heating and recession evidence.

A Executed source listings and reproducible input decks

These listings are actual source and generated MAPDL inputs, not pseudocode. The source archive alongside the report contains complete model decks and configuration files for all four corrected primary runs. Each archived file was checked against its sealed runtime manifest. The main listing uses sustainer Al as the representative 80/20-row case; its complete 909-node/400-element deck is printed at the end. The CF property/orientation block is printed separately with its original line numbers. Booster radii and all case-specific paths are preserved in the accompanying complete input copies.

The listings retain historical names such as `v0` in commands and titles because they are exact copies of the executed inputs. They do not refer to a request to rerun the previous study. The report source archive's `source_manifest.json` records full hashes for all four cases. The material DLL is a compiled artifact already versioned in `model/`; its source and build commands are printed below. Only the existing serial queue controls execution.

A.1 Complete phenolic material definition

All ten temperature points, fifteen user properties and five state variables. This block is also embedded in the complete mesh deck.

Source: `sustainer_al/material_v0.apdl`.

SHA-256: `fc92fb452f2fa170142917a62b5619b7aaf68071b9ba2711129efefb0965887e`.

```
! Phenolic UserMatTh: reference densities; fixed mesh; no thermal expansion.
TB,USER,1,10,15,THERM
TBTEMP,22.0
TBDATA,1,0.25,0.3,900.0,800.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,26.85
TBDATA,1,0.25,0.3,900.0,800.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,326.85
TBDATA,1,0.32,0.35,1100.0,900.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,626.85
TBDATA,1,0.36,0.4,1300.0,1000.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,776.85
TBDATA,1,0.36,0.44,1300.0,1250.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,926.85
TBDATA,1,0.36,0.49,1300.0,1550.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,1226.85
TBDATA,1,0.36,0.68,1300.0,1950.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,1426.85
TBDATA,1,0.36,0.81,1300.0,2060.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,1926.85
TBDATA,1,0.36,1.24,1300.0,2085.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,2476.85
TBDATA,1,0.36,1.73,1300.0,2090.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TB,STATE,1,,5
TBDATA,1,0,0,0,0,0
```

A.2 Complete Fortran UserMatTh routine

The corrected temporal midpoint is explicit in the listing. The routine contains the constitutive response and gas sensible sink; surface consumption and topology changes are in the controller.

Source: `sustainer_al/usermatth.F`.

SHA-256: `218922756fd267532fc6a915d207604b4282e9219cf412cf8af18538a42dc59a`.

```
*deck,usermatth                       USERDISTRIB
      subroutine usermatth(matId, elemId, kDomIntPt, kLayer,
     &                     kSectPt, ldstep, isubst, keycut,
     &                     ncomp, nStatev, nProp, Time, dTime,
     &                     Temp, dTemp, tgrad, ustatev, prop,
     &                     coords, dudt, dudg, flux, dfdt,
     &                     dfdg, cutFactor, hgen, dens,
     &                     var1, var2, var3, var4, var5, var6)
c
c     Phenolic pyrolysis plus local outgassing diagnostics.
c
c     Properties in TB,USER (SI units):
c       1  virgin thermal conductivity       [W/(m K)]
c       2  char thermal conductivity          [W/(m K)]
c       3  virgin specific heat               [J/(kg K)]
c       4  char specific heat                  [J/(kg K)]
c       5  virgin density                      [kg/m3]
c       6  char density                        [kg/m3]
c       7  Arrhenius pre-exponential factor A  [1/s]
c       8  activation energy Ea                [J/mol]
c       9  reaction order n                    [-]
c      10  positive pyrolysis enthalpy         [J/kg]
c      11  Celsius-to-Kelvin offset            [K]
c      12  pyrolysis-gas specific heat         [J/(kg K)]
c      13  gas reference temperature           [K]
c      14  gas sensible-energy fraction        [0..1]
c      15  released-gas yield multiplier       [0..1]
c
c     State variables:
c       1  alpha, 0 = virgin and 1 = char
c       2  alpha rate                          [1/s]
c       3  gas mass source                     [kg/(m3 s)]
c       4  cumulative released-gas density     [kg/m3]
c       5  gas sensible-energy sink magnitude  [W/kg]
c
c     Kinetics:
c       d(alpha)/dt = A exp(-Ea/(R T)) (1-alpha)**n
c
c     The reaction is integrated analytically over the current time
c     increment.  All state is local to an integration point, which
c     keeps the routine safe for shared- and distributed-memory solves.
c
      implicit none

      integer matId, elemId, kDomIntPt, kLayer, kSectPt
      integer ldstep, isubst, keycut, ncomp, nStatev, nProp
      integer i, j

      double precision Time, dTime, Temp, dTemp
      double precision tgrad(ncomp), ustatev(nStatev)
      double precision prop(nProp), coords(3)
      double precision dudt, dudg(ncomp), flux(ncomp)
      double precision dfdt(ncomp), dfdg(ncomp,ncomp)
      double precision cutFactor, hgen, dens
      double precision var1, var2, var3, var4, var5, var6

      double precision gasr, minTemp, maxDAlpha, minCut
      parameter (gasr=8.31446261815324d0)
      parameter (minTemp=1.0d0)
      parameter (maxDAlpha=5.0d-2)
      parameter (minCut=1.0d-1)

      double precision kv, kc, cpv, cpc, rhov, rhoc
      double precision apre, ea, ordern, deltaH, tempOffset
      double precision gasCp, gasTref, gasEnergyFrac, gasYield
      double precision temperatureMid, exponent, rateConstant
```

```
      double precision alphaOld, alphaNew, alphaDot, deltaAlpha
      double precision betaOld, betaNew, power, bracket
      double precision conductivity, specificHeat, densityNew
      double precision gasMassRate, gasMassCumulative, gasHeatSink
      double precision availableGasDensity, gasDeltaTemperature

c     Always initialize every output controlled by UserMatTh.
      keycut = 0
      cutFactor = 1.0d0
      dudt = 1.0d0
      do i = 1, ncomp
         dudg(i) = 0.0d0
         flux(i) = 0.0d0
         dfdt(i) = 0.0d0
         do j = 1, ncomp
            dfdg(i,j) = 0.0d0
         enddo
      enddo

c     A cutback is safer than accessing an incomplete material table.
      if (nProp .lt. 15 .or. nStatev .lt. 5) then
         keycut = 1
         cutFactor = minCut
         return
      endif

      kv            = prop(1)
      kc            = prop(2)
      cpv           = prop(3)
      cpc           = prop(4)
      rhov          = prop(5)
      rhoc          = prop(6)
      apre          = prop(7)
      ea            = prop(8)
      ordern        = prop(9)
      deltaH        = prop(10)
      tempOffset    = prop(11)
      gasCp         = prop(12)
      gasTref       = prop(13)
      gasEnergyFrac = prop(14)
      gasYield      = prop(15)

c     Reject nonphysical values without allowing NaNs into the solve.
      if (kv .le. 0.0d0 .or. kc .le. 0.0d0 .or.
     &    cpv .le. 0.0d0 .or. cpc .le. 0.0d0 .or.
     &    rhov .le. 0.0d0 .or. rhoc .le. 0.0d0 .or.
     &    rhoc .gt. rhov .or. apre .lt. 0.0d0 .or.
     &    ea .lt. 0.0d0 .or. ordern .le. 0.0d0 .or.
     &    deltaH .lt. 0.0d0 .or. gasCp .lt. 0.0d0 .or.
     &    gasTref .lt. 0.0d0 .or. gasEnergyFrac .lt. 0.0d0
     &    .or. gasEnergyFrac .gt. 1.0d0 .or.
     &    gasYield .lt. 0.0d0 .or. gasYield .gt. 1.0d0) then
         keycut = 1
         cutFactor = minCut
         return
      endif

      alphaOld = dmax1(0.0d0, dmin1(1.0d0, ustatev(1)))
c     Temp is the current endpoint; dTemp is endpoint minus start.
c     Subtract half the increment to evaluate the temporal midpoint.
      temperatureMid = Temp - 0.5d0*dTemp + tempOffset
      temperatureMid = dmax1(temperatureMid, minTemp)

      exponent = -ea/(gasr*temperatureMid)
      exponent = dmax1(-700.0d0, dmin1(700.0d0, exponent))
      rateConstant = apre*dexp(exponent)

      alphaNew = alphaOld
      if (dTime .gt. 0.0d0 .and. rateConstant .gt. 0.0d0
     &    .and. alphaOld .lt. 1.0d0) then
         betaOld = dmax1(1.0d0-alphaOld, 0.0d0)
         if (dabs(ordern-1.0d0) .le. 1.0d-10) then
```

```
            betaNew = betaOld*dexp(-rateConstant*dTime)
         else
            power = 1.0d0-ordern
            bracket = betaOld**power
     &                - power*rateConstant*dTime
            if (bracket .le. 0.0d0) then
               betaNew = 0.0d0
            else
               betaNew = bracket**(1.0d0/power)
            endif
         endif
         alphaNew = 1.0d0-betaNew
         alphaNew = dmax1(alphaOld, dmin1(1.0d0, alphaNew))
       endif

       deltaAlpha = alphaNew-alphaOld
       alphaDot = 0.0d0
       if (dTime .gt. 0.0d0) alphaDot = deltaAlpha/dTime

c      Linear virgin/char mixture at the updated conversion.
       conductivity = (1.0d0-alphaNew)*kv + alphaNew*kc
       specificHeat = (1.0d0-alphaNew)*cpv + alphaNew*cpc
       densityNew   = (1.0d0-alphaNew)*rhov + alphaNew*rhoc

c      Released gas is the solid density loss associated with conversion.
       availableGasDensity = gasYield*dmax1(rhov-rhoc, 0.0d0)
       gasMassRate = availableGasDensity*alphaDot
       gasMassCumulative = availableGasDensity*alphaNew
       gasDeltaTemperature = dmax1(temperatureMid-gasTref, 0.0d0)
       gasHeatSink = gasEnergyFrac*gasMassRate*gasCp
      &              *gasDeltaTemperature/dmax1(densityNew, 1.0d-30)

       dudt = specificHeat
       dens = densityNew
       do i = 1, ncomp
          flux(i) = -conductivity*tgrad(i)
          dfdg(i,i) = -conductivity
       enddo

       ustatev(1) = alphaNew
       ustatev(2) = alphaDot
       ustatev(3) = gasMassRate
       ustatev(4) = gasMassCumulative
       ustatev(5) = gasHeatSink

c      For pure thermal elements hgen is heat generation per unit mass.
c      Both positive enthalpy terms are sinks, hence the minus signs.
       hgen = hgen-deltaH*alphaDot-gasHeatSink

c      Limit conversion to five percent per accepted increment.
       if (deltaAlpha .gt. maxDAlpha) then
          keycut = 1
          cutFactor = maxDAlpha/dmax1(deltaAlpha, 1.0d-30)
          cutFactor = dmax1(minCut, dmin1(0.95d0, cutFactor))
       endif

       return
       end
```

A.3 Complete MAPDL transient driver

This driver calls the controller before and after each native solve. The accepted-step gate, rather than a zero solver exit alone, determines whether the next interval may start.

Source: `sustainer_al/run_v0.inp`.

SHA-256: `c729bc3c022295a634e7c2aeeec61a2ca2d59b1d5687806fa4aab54454644397`.

```
! New motor v0; launched only by launch_v0.ps1 -Run.
/INPUT,model_base,inp
/SOLU
ANTYPE,TRANS,NEW
NROPT,FULL
THOPT,FULL
TRNOPT,FULL
TINTP,,,,1
AUTOTS,ON
NEQIT,60
CNVTOL,HEAT,,1e-05,2,1e-6
KBC,1
ESTIF,1E-6
! SMP pilot: no distributed restart option
RESCONTROL,DEFINE,ALL,LAST,-1,,6
OUTRES,ALL,LAST
OUTRES,SVAR,LAST
FINISH
*DO,V0_LOOP,1,100000
  /DELETE,step_control,inp
  /SYS,C:\IrisStudy\runs\sustainer_al_pulse7_n80_midpoint_tight\couple.cmd prepare
  V0_OK=0
  /INPUT,step_control,inp
  *IF,V0_OK,NE,1,THEN
    /COM,V0_PREPARE_FAILED_OR_MISSING
    /EXIT,NOSAVE
  *ENDIF
  *IF,V0_DONE,EQ,1,THEN
    *EXIT
  *ENDIF
  /SOLU
  *IF,V0_INDEX,GT,1,THEN
    ANTYPE,,REST,,,CONTINUE
  *ENDIF
  /INPUT,step_control,inp
  /INPUT,apply_loads,inp
  TIME,V0_TARGET
  DELTIM,V0_DT,0.0005,V0_DT,OFF
  SOLVE
  /INPUT,export_fields,mac
  /DELETE,accepted,inp
  /SYS,C:\IrisStudy\runs\sustainer_al_pulse7_n80_midpoint_tight\couple.cmd accept
  V0_OK=0
  /INPUT,accepted,inp
  *IF,V0_OK,NE,1,THEN
    /COM,V0_POSTPROCESS_OR_BALANCE_FAILED
    /EXIT,NOSAVE
  *ENDIF
*ENDDO
/COM,V0_CONTROLLER_STOPPED_CHECK_STATE_JSON
/EXIT,NOSAVE
```

A.4 Complete MAPDL resume driver

Only a matching accepted native/controller checkpoint is a valid restart. A restart does not authorize pairing an earlier controller record with a later native state.

Source: `sustainer_al/resume_v0.inp`.

SHA-256: `2c0e81d2a6393a15dfa749c9cdbb32a9789f3c84dff55362a20cddc80144eb25`.

```
! Resume only from a verified accepted checkpoint.
*DO,V0_LOOP,1,100000
  /DELETE,step_control,inp
  /SYS,C:\IrisStudy\runs\sustainer_al_pulse7_n80_midpoint_tight\couple.cmd prepare
  V0_OK=0
  /INPUT,step_control,inp
  *IF,V0_OK,NE,1,THEN
    /COM,V0_PREPARE_FAILED_OR_MISSING
    /EXIT,NOSAVE
  *ENDIF
  *IF,V0_DONE,EQ,1,THEN
    *EXIT
  *ENDIF
  /SOLU
  *IF,V0_INDEX,GT,1,THEN
    ANTYPE,,REST,V0_PREVIOUS,,CONTINUE
  *ENDIF
  /INPUT,step_control,inp
  /INPUT,apply_loads,inp
  TIME,V0_TARGET
  DELTIM,V0_DT,0.0005,V0_DT,OFF
  SOLVE
  /INPUT,export_fields,mac
  /DELETE,accepted,inp
  /SYS,C:\IrisStudy\runs\sustainer_al_pulse7_n80_midpoint_tight\couple.cmd accept
  V0_OK=0
  /INPUT,accepted,inp
  *IF,V0_OK,NE,1,THEN
    /COM,V0_POSTPROCESS_OR_BALANCE_FAILED
    /EXIT,NOSAVE
  *ENDIF
*ENDDO
/COM,V0_CONTROLLER_STOPPED_CHECK_STATE_JSON
/EXIT,NOSAVE
```

A.5 Complete MAPDL field and energy exporter

The exporter records actual native time and substep identity, retrieves defined live phenolic state variables, and keeps raw energy channels for diagnosis.

Source: `sustainer_al/export_fields.mac`.

SHA-256: `f3265b15f2989307d8b1869889c84a2a71117955465659daa20b5baead815072`.

```
FINISH
/POST1
SET,LAST
ALLSEL,ALL
*GET,V0_TIME,ACTIVE,0,SET,TIME
*CFOPEN,observed_time,txt
*VWRITE,V0_TIME
(E24.16)
*CFCLOSE
*GET,V0_LSTEP,ACTIVE,0,SET,LSTP
*GET,V0_SUBSTEP,ACTIVE,0,SET,SBST
*CFOPEN,solver_step_identity,txt
*VWRITE,V0_LSTEP,V0_SUBSTEP,V0_TIME
(2(F12.0,1X),E24.16)
*CFCLOSE
*DEL,V0_N
*DIM,V0_N,ARRAY,909
*DEL,V0_T
*DIM,V0_T,ARRAY,909
*DEL,V0_M
*DIM,V0_M,ARRAY,909
*DEL,V0_E
*DIM,V0_E,ARRAY,400
*VFILL,V0_N(1),RAMP,1,1
*VGET,V0_T(1),NODE,1,TEMP
*VGET,V0_M(1),NODE,1,NSEL
*CFOPEN,node_temperatures,txt
*VMASK,V0_M(1)
*VWRITE,V0_N(1),V0_T(1)
(F10.0,1X,E24.16)
*CFCLOSE
*VFILL,V0_E(1),RAMP,1,1
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,ERAS
*DEL,V0_ALP
*DIM,V0_ALP,ARRAY,400
ESEL,S,MAT,,1
ESEL,R,LIVE
ETABLE,ALP,SVAR,1
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_ALP(V0_FIRST),ELEM,V0_FIRST,ETAB,ALP
*VLEN
*DEL,V0_GAS
*DIM,V0_GAS,ARRAY,400
ESEL,S,MAT,,1
ESEL,R,LIVE
ETABLE,GAS,SVAR,3
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_GAS(V0_FIRST),ELEM,V0_FIRST,ETAB,GAS
*VLEN
*DEL,V0_GSEN
*DIM,V0_GSEN,ARRAY,400
ESEL,S,MAT,,1
ESEL,R,LIVE
ETABLE,GSEN,SVAR,5
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_GSEN(V0_FIRST),ELEM,V0_FIRST,ETAB,GSEN
*VLEN
*DEL,V0_QZ
```

```
*DIM,V0_QZ,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,QZ,TF,Z
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_QZ(V0_FIRST),ELEM,V0_FIRST,ETAB,QZ
*VLEN
*DEL,V0_CAP
*DIM,V0_CAP,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,CAP,NMISC,38
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_CAP(V0_FIRST),ELEM,V0_FIRST,ETAB,CAP
*VLEN
*DEL,V0_GEN
*DIM,V0_GEN,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,GEN,NMISC,39
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_GEN(V0_FIRST),ELEM,V0_FIRST,ETAB,GEN
*VLEN
*DEL,V0_CONV
*DIM,V0_CONV,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,CONV,NMISC,40
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_CONV(V0_FIRST),ELEM,V0_FIRST,ETAB,CONV
*VLEN
*DEL,V0_HFX
*DIM,V0_HFX,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,HFX,NMISC,41
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_HFX(V0_FIRST),ELEM,V0_FIRST,ETAB,HFX
*VLEN
*DEL,V0_RAD
*DIM,V0_RAD,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,RAD,NMISC,42
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_RAD(V0_FIRST),ELEM,V0_FIRST,ETAB,RAD
*VLEN
*DEL,V0_HCONV
*DIM,V0_HCONV,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
ETABLE,HCONV,NMISC,29
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_HCONV(V0_FIRST),ELEM,V0_FIRST,ETAB,HCONV
*VLEN
*DEL,V0_OCONV
*DIM,V0_OCONV,ARRAY,400
ESEL,S,TYPE,,1,2
```

```
ESEL,R,LIVE
ETABLE,OCONV,NMISC,17
*GET,V0_FIRST,ELEM,0,NUM,MIN
*GET,V0_COUNT,ELEM,0,COUNT
*VLEN,V0_COUNT
*VGET,V0_OCONV(V0_FIRST),ELEM,V0_FIRST,ETAB,OCONV
*VLEN
*DEL,V0_EM
*DIM,V0_EM,ARRAY,400
ESEL,S,TYPE,,1,2
ESEL,R,LIVE
*VGET,V0_EM(1),ELEM,1,ESEL
*CFOPEN,element_fields,txt
*VMASK,V0_EM(1)
*VWRITE,V0_E(1),V0_ALP(1),V0_GAS(1),V0_GSEN(1),V0_QZ(1)
(F10.0,4(1X,E24.16))
*CFCLOSE
*CFOPEN,energy_fields,txt
*VMASK,V0_EM(1)
*VWRITE,V0_E(1),V0_CAP(1),V0_GEN(1),V0_CONV(1),V0_HFX(1),V0_RAD(1),V0_HCONV(1),V0_OCONV(1)
(F10.0,7(1X,E22.14))
*CFCLOSE
ALLSEL,ALL
FINISH
```

A.6 CF material and local-coordinate block

This is the complete CF property/orientation definition extracted from its larger mesh deck. Line numbers refer to the original deck. The first following element command and later connectivity are in the archived full file.
Source: `sustainer_cf/model_base.inp`.
SHA-256: `d20f8f3455f68bd5c1a0b8ac036ce6bfe0def16cbd8739918b7523724a6a5cf2`.

```
MPDELE,ALL,2
MP,DENS,2,1451
MPTEMP,1,22
MPDATA,KXX,2,1,2.5
MPDATA,KYY,2,1,2.5
MPDATA,KZZ,2,1,0.55
MPDATA,C,2,1,750.6106
MPTEMP,2,100
MPDATA,KXX,2,2,2.5
MPDATA,KYY,2,2,2.5
MPDATA,KZZ,2,2,0.55
MPDATA,C,2,2,975.04
MPTEMP,3,150
MPDATA,KXX,2,3,2.5
MPDATA,KYY,2,3,2.5
MPDATA,KZZ,2,3,0.55
MPDATA,C,2,3,1118.905
MPTEMP,4,200
MPDATA,KXX,2,4,2.5
MPDATA,KYY,2,4,2.5
MPDATA,KZZ,2,4,0.55
MPDATA,C,2,4,1262.77
MPTEMP,5,300
MPDATA,KXX,2,5,2.5
MPDATA,KYY,2,5,2.5
MPDATA,KZZ,2,5,0.55
MPDATA,C,2,5,1262.77
MPTEMP,6,600
MPDATA,KXX,2,6,2.5
MPDATA,KYY,2,6,2.5
MPDATA,KZZ,2,6,0.55
MPDATA,C,2,6,1262.77
MPTEMP,7,1000
MPDATA,KXX,2,7,2.5
MPDATA,KYY,2,7,2.5
MPDATA,KZZ,2,7,0.55
MPDATA,C,2,7,1262.77
MPTEMP,8,1500
MPDATA,KXX,2,8,2.5
MPDATA,KYY,2,8,2.5
MPDATA,KZZ,2,8,0.55
MPDATA,C,2,8,1262.77
MPTEMP,9,2230.85
MPDATA,KXX,2,9,2.5
MPDATA,KYY,2,9,2.5
MPDATA,KZZ,2,9,0.55
MPDATA,C,2,9,1262.77
K,100001,0,0,0
K,100002,0.9999976201773518,0,0.0021816598343367697
K,100003,0,1,0
CSKP,11,0,100001,100002,100003
KDELE,100001,100003
K,100001,0,0,0
K,100002,0.9999976201773518,0,-0.00218165983433677
K,100003,0,1,0
CSKP,12,0,100001,100002,100003
KDELE,100001,100003
CSYS,0
ESYS,11
```

A.7 Complete surface and acceptance controller

Included because the Fortran routine alone does not perform ablation. This controller supplies the explicit surface law, blowing, removal scheduling, balance checks and accepted-state persistence.
Source: `sustainer_al/coupler.py`.
SHA-256: `1bde9f56c22f19aa00f7ba1ebb433f9ad424a5ae67f9f591476587ce7b2b8c69`.

```
"""Partitioned thermal/char coupling, called by MAPDL only between solves.

Standard library only. All runtime files belong to this iteration. `prepare`
does not solve anything. `accept` checks actual converged MAPDL field exports.
"""
from pathlib import Path
import argparse
import csv
import json
import math
import os
import traceback
import hashlib
import sys

SIGMA = 5.670374419e-8
R = 8.314462618


def atomic(path, value):
    path=Path(path)
    temporary=path.with_suffix(path.suffix+'.new')
    temporary.write_text(value,encoding='utf-8')
    os.replace(temporary,path)


def load(path):
    return json.loads(Path(path).read_text(encoding='utf-8-sig'))


def save(path,value):
    atomic(path,json.dumps(value,indent=2,allow_nan=False))


def fingerprint(c):
    return hashlib.sha256(json.dumps(c,sort_keys=True).encode()).hexdigest()


def validate_config(c):
    assert c['qualification']=='EXPLORATORY_NOT_CALIBRATED' or c['char_law']['calibrated']
    assert c['end_time_s']>0 and 0<c['minimum_dt_s']<=c['macro_dt_s']
    assert 1 <= c['ranks'] <= 14
    g=c['geometry']
    assert g['phenolic_radial_rows']>0 and g['faces_per_row']>0
    assert math.isclose(g['phenolic_outer_radius_m']-g['inner_radius_m'],
                        g['phenolic_thickness_m'],rel_tol=0,abs_tol=1e-12)
    assert math.isclose(g['outer_radius_m']-g['phenolic_outer_radius_m'],
                        g['aluminium_thickness_m'],rel_tol=0,abs_tol=1e-12)
    m=c['material']; law=c['char_law']; h=c['heating']; n=c['numerics']
    assert 0<m['char_reference_density_kg_m3']<m['virgin_reference_density_kg_m3']
    assert m.get('phenolic_k_scale',1.)>0 and m.get('phenolic_cp_scale',1.)>0
    assert law['effective_endothermic_cost_J_kg']>0
    assert law['mode'] in ('effective_exploratory','species')
    assert law['pressure_Pa'] is None or law['pressure_Pa']>=0
    for key in ('exploratory_pressure_ratio','reactive_activity','reference_rate_kg_m2_s',
                'activation_energy_J_mol','pressure_exponent','mass_transfer_cap_kg_m2_s'):
        assert law[key]>=0,key
    assert law['reference_temperature_K']>0 and law['pressure_reference_Pa']>0
    assert 0<n['row_residual_mass_fraction']<.01
    assert 0<n['max_char_fraction_per_step']<=.25
    assert h['gas_temperature_C']>-273.15 and h['h0_W_m2_K']>0
    assert 0<=h['hot_effective_emissivity']<=1 and 0<=h['outer_emissivity']<=1
    if law['mode']=='species':
        assert law['pressure_Pa'] is not None,'Species model needs an explicit chamber pressure.'
        assert law['species'],'Species model needs supplied reactant mole fractions and kinetics.'
        assert sum(x['mole_fraction'] for x in law['species'])<=1+1e-12
```

```
        for s in law['species']:
            assert s['name'] in ('H2O','CO2','O2') and 0<=s['mole_fraction']<=1
            for key in ('reference_rate_kg_m2_s','activation_energy_J_mol',
                        'pressure_exponent','mass_transfer_cap_kg_m2_s'):
                assert s[key]>=0


def char_rate(T_C,law):
    """Carbon consumption flux, kg m^-2 s^-1; no conversion cutoff at 0.99.

    Each effective channel combines kinetics and mass-transfer resistance.
    Positive H_eff represents an endothermic effective removal cost only.
    Species mode is a configurable engineering closure, not equilibrium chemistry.
    """
    T=T_C+273.15
    if T<=0: raise ValueError('Nonphysical absolute temperature')
    p=law['pressure_Pa']
    ratio=p/law['pressure_reference_Pa'] if p is not None else law['exploratory_pressure_ratio']
    channels=law['species'] if law['mode']=='species' else [law]
    total=0.0
    for s in channels:
        activity=s.get('mole_fraction',law['reactive_activity'])
        if ratio<=0 or activity<=0: continue
        partial=ratio*activity
        exponent=-s['activation_energy_J_mol']/R*(1/T-1/law['reference_temperature_K'])
        kinetics=s['reference_rate_kg_m2_s']*math.exp(max(-700,min(100,exponent)))*partial**s['pressure_exponent
    ']
        transport=s['mass_transfer_cap_kg_m2_s']*partial
        if kinetics>0 and transport>0:
            total+=1/(1/kinetics+1/transport)
    return total


def blowing(h0,gflux,cp):
    b=max(gflux,0)*cp/h0
    if b<1e-7: return h0*(1-b/2+b*b/12)
    if b>700: return 0.0
    return h0*b/math.expm1(b)


def face_temperature(f,temperatures,initial):
    return sum(temperatures.get(str(n),initial) for n in f['nodes'])/4


def initial_state(mesh,c):
    rho=c['material']['virgin_reference_density_kg_m3']
    return dict(config_sha256=fingerprint(c),index=0,time_s=0.0,layer=0,alpha={},temperatures={},char_consumed
    ={},
                mass_initial_kg=sum(e['volume']*rho for e in mesh['elements'].values() if e['mat']==1),
                gas_cumulative_kg=0.0,char_cumulative_kg=0.0,residual_ejected_kg=0.0,
                removed_mesh_energy_J=0.0,energy_integrals={},gas_rate_kg_s=0.0,
                radial_flux={},status='PREPARED_NOT_STARTED',pending_kills=[],warnings=[])


def exposure(c,time):
    h=dict(c['heating'])
    on=time<h.get('pulse_duration_s',float('inf'))-1e-10
    if not on:
        h.update(h0_W_m2_K=h.get('post_pulse_h_W_m2_K',8.),
                 gas_temperature_C=h['ambient_temperature_C'],hot_effective_emissivity=0.)
    h['exposure_on']=on
    return h


def prepare_step(mesh,c,s):
    h=exposure(c,s['time_s']); m=c['material']; n=c['numerics']
    layer=s['layer']; row=mesh['rows'][layer]
    area=sum(f['area'] for f in row)
    hfilm=blowing(h['h0_W_m2_K'],(s['gas_rate_kg_s']+s.get('char_rate_kg_s',0.))/area,h['gas_cp_J_kg_K']) if h['
    exposure_on'] else h['h0_W_m2_K']
    dt=min(c['macro_dt_s'] if h['exposure_on'] else c.get('post_pulse_macro_dt_s',c['macro_dt_s']),c['end_time_s
    ']-s['time_s'])
```

```
    if h['exposure_on']:dt=min(dt,h.get('pulse_duration_s',float('inf'))-s['time_s'])
    elif c.get('post_pulse_initial_duration_s',0)>0:
        initial_end=h['pulse_duration_s']+c['post_pulse_initial_duration_s']
        if s['time_s']<initial_end-1e-10:
            dt=min(dt,c['post_pulse_initial_dt_s'],initial_end-s['time_s'])
    commands=['ALLSEL,ALL','FDELE,ALL,HEAT','ESEL,S,TYPE,,1,4',
              'SFEDELE,ALL,ALL,CONV','ALLSEL,ALL']
    # Multiframe restart restores the initial .rdb; .rnnn does not retain
    # EKILL/EALIVE. Reissue the entire removed prefix after EVERY restart.
    # See the v261 Basic Analysis Guide, Multiframe Restart Limitations.
    for removed_row in mesh['rows'][:layer]:
        for f in removed_row: commands.append(f'EKILL,{f["element"]}')
    commands+=['ESEL,S,TYPE,,1,4','ESEL,R,LIVE']
    expected_live=len(mesh['elements'])-sum(len(r) for r in mesh['rows'][:layer])
    commands += ['*GET,V0_LIVE,ELEM,0,COUNT',
                 '*CFOPEN,live_topology_before_solve,txt','*VWRITE,V0_LIVE',
                 '(F12.0)','*CFCLOSE',f'*IF,V0_LIVE,NE,{expected_live},THEN',
                 '/COM,IRIS_LIVE_TOPOLOGY_MISMATCH','/EXIT,NOSAVE','*ENDIF']
    forces={}; faces=[]
    totals=dict(hot_radiation_W=0.,outer_radiation_W=0.,ablation_sink_W=0.,char_rate_kg_s=0.)
    for f in row:
        eid=str(f['element']); T=face_temperature(f,s['temperatures'],c['initial_temperature_C'])
        alpha=s['alpha'].get(eid,0.)
        consumed=s['char_consumed'].get(eid,0.)
        char_total=m['char_reference_density_kg_m3']*f['volume']
        char_available=max(alpha*char_total-consumed,0.)
        qconv=hfilm*(h['gas_temperature_C']-T)
        qrad=h['hot_effective_emissivity']*SIGMA*((h['radiative_temperature_C']+273.15)**4-(T+273.15)**4)
        # Conduction estimate comes from the previous converged row flux.
        # Pyrolysis and gas sensible sinks already act inside UserMatTh.
        qcond=s['radial_flux'].get(eid,0.)
        energy_cap=max(qconv+qrad-qcond,0.)/c['char_law']['effective_endothermic_cost_J_kg']
        flux=min(alpha*char_rate(T,c['char_law']),energy_cap,
                 char_available/(dt*f['area']),n['max_char_fraction_per_step']*char_total/(dt*f['area'])) if h['
    exposure_on'] else 0.
        char_power=flux*f['area']*c['char_law']['effective_endothermic_cost_J_kg']
        extra=qrad*f['area']-char_power
        for nid in f['nodes']: forces[nid]=forces.get(nid,0.)+extra/4
        commands += [f'SFE,{f["element"]},{f["face"]},CONV,0,{hfilm:.16e}',
                     f'SFE,{f["element"]},{f["face"]},CONV,2,{h["gas_temperature_C"]:.16e}']
        faces.append(dict(element=f['element'],area=f['area'],T_C=T,h=hfilm,Tgas_C=h['gas_temperature_C'],
                          char_rate=flux*f['area'],char_available=char_available,
                          qconv=qconv,qrad=qrad,qcond_est=qcond))
        totals['hot_radiation_W']+=qrad*f['area']
        totals['ablation_sink_W']+=char_power
        totals['char_rate_kg_s']+=flux*f['area']
    for f in mesh['outer']:
        T=face_temperature(f,s['temperatures'],c['initial_temperature_C'])
        qrad=h['outer_emissivity']*SIGMA*((h['ambient_temperature_C']+273.15)**4-(T+273.15)**4)
        for nid in f['nodes']: forces[nid]=forces.get(nid,0.)+qrad*f['area']/4
        commands += [f'SFE,{f["element"]},{f["face"]},CONV,0,{h["outer_h_W_m2_K"]:.16e}',
                     f'SFE,{f["element"]},{f["face"]},CONV,2,{h["ambient_temperature_C"]:.16e}']
        totals['outer_radiation_W']+=qrad*f['area']
    commands+=['ALLSEL,ALL']+[f'F,{nid},HEAT,{val:.16e}' for nid,val in sorted(forces.items())]
    commands+=['ALLSEL,ALL']
    p=dict(index=s['index']+1,layer=layer,start=s['time_s'],target=s['time_s']+dt,dt=dt,
           area=area,faces=faces,forces=forces,totals=totals,
           exposure_on=h['exposure_on'],loaded_hot_faces=len(row),loaded_outer_faces=len(mesh['outer']),
           loaded_hot_nodes=len(set(n for f in row for n in f['nodes'])))
    expected_nodes=(c['geometry']['axial_divisions']+1)*(c['geometry']['angular_divisions']+1)
    assert p['loaded_hot_faces']==c['geometry']['faces_per_row']
    assert p['loaded_hot_nodes']==expected_nodes
    assert math.isclose(sum(forces.values()),totals['hot_radiation_W']+totals['outer_radiation_W']-totals['
    ablation_sink_W'],abs_tol=1e-10)
    return p,'\n'.join(commands)+'\n'


def read_rows(path):
    values={}
    for line in Path(path).read_text().splitlines():
        p=line.split()
        if not p: continue
```

```
        nums=[float(x.replace('D','E')) for x in p]
        if not all(math.isfinite(x) for x in nums): raise ValueError('Nonfinite solver output')
        key=str(int(nums[0]))
        if key in values: raise ValueError('Duplicate solver output ID '+key)
        values[key]=nums[1:]
    return values


def preserve_accepted_restart(root,state):
    """Keep two verified accepted native generations outside MAPDL's ring.

    Repeated multiframe restores can overwrite the same native .rnnn slot.
    Snapshot essential native files before the next solve can replace them.
    A manifest written last makes an interrupted copy detectably incomplete;
    the alternate slot still retains the preceding accepted generation.
    """
    import re, shutil
    root=Path(root)
    files=sorted(p for p in root.iterdir() if p.is_file() and
                 (p.name in ('file.rdb','file.ldhi') or re.fullmatch(r'file\.r\d{3}',p.name)))
    names={p.name for p in files}
    if not {'file.rdb','file.ldhi'}.issubset(names) or not any(re.fullmatch(r'file\.r\d{3}',n) for n in names):
        raise ValueError('Native restart files missing at accepted time')
    slot=root/f'accepted_restart_{state["index"]%2}'
    slot.mkdir(exist_ok=True)
    hashes={}
    for src in files:
        target=slot/src.name;temporary=target.with_suffix(target.suffix+'.new')
        shutil.copy2(src,temporary)
        digest=hashlib.sha256(src.read_bytes()).hexdigest()
        if hashlib.sha256(temporary.read_bytes()).hexdigest()!=digest:
            raise ValueError('Accepted restart copy hash mismatch')
        os.replace(temporary,target);hashes[src.name]=digest
    save(slot/'state.json',state)
    hashes['state.json']=hashlib.sha256((slot/'state.json').read_bytes()).hexdigest()
    save(slot/'manifest.json',dict(accepted_index=state['index'],accepted_time_s=state['time_s'],
         config_sha256=state['config_sha256'],files=hashes))


def validate_native_step_identity(values,pending):
    """Endpoint-only heat-rate exports support exactly one native substep.

    Do not combine a final internal-substep capacity/gas rate with a full
    macro-interval conversion increment. A cutback must stop acceptance.
    """
    loadstep,substep,time=values
    if loadstep!=pending['index'] or abs(time-pending['target'])>1e-8:
        raise ValueError('Native load-step identity or time mismatch')
    if substep!=1:
        raise ValueError('Internal substepping is unsupported by the endpoint macro energy audit; preserve the
   preceding accepted restart and refine the controller interval')


def accept_step(mesh,c,s,p,temperatures,fields,energy,time):
    if abs(time-p['target'])>1e-8: raise ValueError('Solved time does not match target')
    if time<=s['time_s']: raise ValueError('Solved time did not advance')
    m=c['material']; n=c['numerics']; rv=m['virgin_reference_density_kg_m3']; rc=m['char_reference_density_kg_m3
   ']
    # POST1 in MAPDL 2026 R1 keeps result rows for EKILLed elements even after
    # ESEL,R,LIVE.  Treat those rows as stale output, not as a resurrection:
    # require every live solid exactly once, allow only controller-declared dead
    # solids as extras, then exclude the dead rows from all state and audits.
    dead={str(f['element']) for row in mesh['rows'][:p['layer']] for f in row}
    expected=set(mesh['elements'])-dead
    field_ids=set(fields); energy_ids=set(energy); defined=set(mesh['elements'])
    if field_ids!=energy_ids:
        raise ValueError('Element and energy field ID sets differ')
    if not expected.issubset(field_ids):
        raise ValueError('Missing live solid fields')
    if not field_ids.issubset(defined) or not (field_ids-expected).issubset(dead):
        raise ValueError('Unexpected solid field IDs')
    fields={eid:fields[eid] for eid in expected}
```

```
    energy={eid:energy[eid] for eid in expected}
    if any(len(v)!=4 for v in fields.values()) or any(len(v)!=7 for v in energy.values()):
        raise ValueError('Truncated solver fields')
    used_nodes={str(nid) for f in mesh['rows'][p['layer']]+mesh['outer'] for nid in f['nodes']}
    if not used_nodes.issubset(temperatures): raise ValueError('Missing exposed-face nodal temperatures')
    s=json.loads(json.dumps(s))
    pyro_power=gas_power=0.; gas_delta=0.; live_mesh_mass=0.
    for eid,e in mesh['elements'].items():
        if e['mat']!=1 or eid in dead: continue
        a,gas,gsens,qz=fields[eid]
        old=s['alpha'].get(eid,0.)
        if a<-1e-8 or a>1+1e-8 or a<old-1e-7: raise ValueError('Invalid/descending pyrolysis state '+eid)
        a=min(1.,max(0.,a)); rho=rv-(rv-rc)*a
        gas_delta+=(rv-rc)*(a-old)*e['volume']
        pyro_power+=rho*m['pyrolysis_enthalpy_J_kg']*(a-old)/p['dt']*e['volume']
        gas_power+=rho*gsens*e['volume']
        live_mesh_mass+=rho*e['volume']
        s['alpha'][eid]=a
        s['radial_flux'][eid]=qz
    for f in p['faces']:
        eid=str(f['element']); delta=f['char_rate']*p['dt']
        s['char_consumed'][eid]=s['char_consumed'].get(eid,0.)+delta
        allowed=rc*mesh['elements'][eid]['volume']*s['alpha'][eid]
        if s['char_consumed'][eid]>allowed+1e-14: raise ValueError('Carbon consumption exceeds produced char')
    s['gas_cumulative_kg']+=gas_delta
    s['char_cumulative_kg']+=p['totals']['char_rate_kg_s']*p['dt']
    pending_char=sum(s['char_consumed'].get(eid,0.) for eid in expected if mesh['elements'][eid]['mat']==1)
    remaining=live_mesh_mass-pending_char
    mass_residual=s['mass_initial_kg']-remaining-s['gas_cumulative_kg']-s['char_cumulative_kg']-s['
residual_ejected_kg']
    if abs(mass_residual)>max(1e-13,1e-6*s['mass_initial_kg']): raise ValueError('Phe0 mass balance not closed')
    cap,gen_solver,conv_solver,hfx,rad,hot_face_conv,outer_face_conv=[sum(v[i] for v in energy.values()) for i in
     range(7)]
    nodal=sum(float(x) for x in p['forces'].values())
    # In MAPDL 2026 R1, SOLID278/NMISC,39 remains zero for the hgen sink
    # supplied by UserMatTh.  The same two internal sinks are independently
    # reconstructed from the accepted conversion increment and SVAR 5, so use
    # their negative sum in the global audit and retain the raw solver channel
    # as a diagnostic.
    gen=-(pyro_power+gas_power)
    # SOLID278/NMISC,40 under-reports the applied convection when UserMatTh is
    # active.  The documented face heat rates (face 5 hot side, face 3 outer
    # side) reproduce the SFE loads, so use their sum in the balance and keep
    # the aggregate channel only as a diagnostic.
    conv=hot_face_conv+outer_face_conv
    energy_residual=cap+conv+rad-hfx-gen-nodal
    scale=max(abs(cap),abs(conv)+abs(rad),abs(hfx)+abs(gen)+abs(nodal),1e-6)
    erel=abs(energy_residual)/scale
    row=mesh['rows'][p['layer']]
    hot_actual=-sum(energy[str(f['element'])][5] for f in row)
    hot_expected=sum(f['area']*p['faces'][i]['h']*(p['faces'][i].get('Tgas_C',c['heating']['gas_temperature_C'])-
                     face_temperature(f,temperatures,c['initial_temperature_C'])) for i,f in enumerate(row))
    hrel=abs(hot_actual-hot_expected)/max(abs(hot_expected),1e-6)
    if hrel>n['hot_power_relative_tolerance']: raise ValueError('Actual hot-side convection absent or mismatched:
     '+str(hrel))
    if erel>n['energy_relative_tolerance'] and n['stop_on_balance_error']:
        raise ValueError('Thermal energy audit failed: '+str(erel))
    s['temperatures']=temperatures
    s['gas_rate_kg_s']=gas_delta/p['dt']
    s['char_rate_kg_s']=p['totals']['char_rate_kg_s']
    tchange=max(abs(face_temperature(f,temperatures,c['initial_temperature_C'])-p['faces'][i]['T_C']) for i,f in
    enumerate(row))
    warnings=[]
    if tchange>n['coupling_temperature_warning_K']: warnings.append('COUPLING_DT_REFINEMENT_REQUIRED')
    # Quantify storage removed by topology. Its energy leaves with the killed
    # FE cell; it is not silently counted as an ablation heat sink again.
    row_remaining=sum((rv-(rv-rc)*s['alpha'][str(f['element'])])*f['volume']-
                      s['char_consumed'].get(str(f['element']),0.) for f in row)
    row_initial=rv*sum(f['volume'] for f in row)
    s['pending_kills']=[]
    killed=0
    if row_remaining<=n['row_residual_mass_fraction']*row_initial:
```

```
        s['pending_kills']=[f['element'] for f in row]
        s['residual_ejected_kg']+=max(row_remaining,0.)
        s['layer']+=1; killed=len(row)
        # A quadrature diagnostic, not a conservative enthalpy remap. The
        # legacy FE law uses cp(T,alpha)*dT, not d[cp(T,alpha)*T].
        for f in row:
            a=s['alpha'][str(f['element'])]
            T=sum(temperatures[str(nid)] for nid in mesh['elements'][str(f['element'])]['nodes'])/8
            hv,hc=sensible_enthalpy(T,c['initial_temperature_C'])
            hv*=m.get('phenolic_cp_scale',1.);hc*=m.get('phenolic_cp_scale',1.)
            s['removed_mesh_energy_J']+=(rv-(rv-rc)*a)*f['volume']*((1-a)*hv+a*hc)
    s['index']=p['index']; s['time_s']=time
    nrows=c['geometry']['phenolic_radial_rows']
    dr=c['geometry']['phenolic_thickness_m']/nrows
    scale=c['geometry']['full_ring_scale_factor']
    s['status']='PHENOLIC_CONSUMED' if s['layer']==nrows else ('COMPLETED_HORIZON' if time>=c['end_time_s']-1e-10
    else 'RUNNING')
    s['warnings']=warnings
    for name,value in dict(storage_W=cap,body_source_W=gen,convection_out_W=conv,
                           nodal_net_W=nodal,ablation_sink_W=p['totals']['ablation_sink_W'],
                           hot_radiation_W=p['totals']['hot_radiation_W'],
                           outer_radiation_W=p['totals']['outer_radiation_W'],
                           pyrolysis_sink_est_W=pyro_power,gas_sensible_sink_est_W=gas_power).items():
        s['energy_integrals'][name.replace('_W','_J')]=s['energy_integrals'].get(name.replace('_W','_J'),0.)+
    value*p['dt']
    removed_initial_mass=sum(
        m['virgin_reference_density_kg_m3']*f['volume']
        for deleted_row in mesh['rows'][:s['layer']] for f in deleted_row)
    alpha_values=[s['alpha'][str(f['element'])] for f in row]
    record=dict(time_s=time,dt_s=p['dt'],layer_solved=p['layer'],rows_removed=p['layer'],rows_scheduled=s['layer
    '],
                hot_radius_m=c['geometry']['inner_radius_m']+p['layer']*dr,
                phenolic_initial_thickness_m=c['geometry']['phenolic_thickness_m'],
                phenolic_removed_thickness_m=s['layer']*dr,
                phenolic_remaining_thickness_m=max(c['geometry']['phenolic_thickness_m']-s['layer']*dr,0.0),
                phenolic_removed_initial_mass_sector_kg=removed_initial_mass,
                phenolic_removed_initial_mass_full_ring_kg=removed_initial_mass*scale,
                hot_faces=p['loaded_hot_faces'],hot_nodes=p['loaded_hot_nodes'],area_m2=p['area'],
                Ts_mean_C=sum(f['area']*face_temperature(f,temperatures,c['initial_temperature_C']) for f in row)
    /p['area'],
                alpha_mean=sum(s['alpha'][str(f['element'])]*f['volume'] for f in row)/sum(f['volume'] for f in
    row),
                alpha_min=min(alpha_values),
                h_W_m2_K=p['faces'][0]['h'],gas_temperature_C=p['faces'][0].get('Tgas_C',c['heating']['
    gas_temperature_C']),
                exposure_on=p.get('exposure_on',True),hot_power_expected_W=hot_expected,hot_power_actual_W=
    hot_actual,
                hot_power_relative_error=hrel,
                hot_power_actual_convrate_W=-sum(energy[str(f['element'])][2] for f in row),
                storage_W=cap,body_generation_W=gen,
                body_generation_solver_W=gen_solver,
                pyrolysis_sink_W=pyro_power,gas_sensible_sink_W=gas_power,
                convection_out_W=conv,convection_solver_W=conv_solver,
                nodal_net_W=nodal,ablation_sink_W=p['totals']['ablation_sink_W'],
                hot_radiation_W=p['totals']['hot_radiation_W'],outer_radiation_W=p['totals']['outer_radiation_W
    '],
                energy_residual_W=energy_residual,energy_relative_error=erel,
                gas_cumulative_kg=s['gas_cumulative_kg'],gas_cumulative_full_ring_kg=s['gas_cumulative_kg']*scale
    ,
                char_cumulative_kg=s['char_cumulative_kg'],char_cumulative_full_ring_kg=s['char_cumulative_kg']*
    scale,
                solid_remaining_kg=max(remaining-row_remaining,0.) if killed else remaining,
                solid_remaining_full_ring_kg=(max(remaining-row_remaining,0.) if killed else remaining)*scale,
                subcell_mass_still_in_FE_kg=pending_char,residual_ejected_kg=s['residual_ejected_kg'],
                mass_residual_kg=mass_residual,elements_scheduled_for_removal=killed,
                topology_sensible_energy_scheduled_est_J=s['removed_mesh_energy_J'],
                max_coupling_temperature_change_K=tchange,status=s['status'],warning=';'.join(warnings))
    for label, ids in mesh.get('temperature_nodes',{}).items():
        record[label+'_max_C'] = max(temperatures[str(i)] for i in ids)
        record[label+'_mean_C'] = sum(temperatures[str(i)] for i in ids)/len(ids)
    record['case_property_range_exceeded'] = record.get('case_inner_max_C',0) > c.get('case_property_limit_C',1e9
    )
```

```
    if record['case_property_range_exceeded'] and c.get('stop_at_case_property_limit',False):
        s['status']='CASE_PROPERTY_LIMIT_REACHED';record['status']=s['status']
    return s,record


def heat_capacity(T):
    table=[(26.85,900,800),(326.85,1100,900),(626.85,1300,1000),
           (776.85,1300,1250),(926.85,1300,1550),(1226.85,1300,1950),
           (1426.85,1300,2060),(1926.85,1300,2085),(2476.85,1300,2090)]
    if T<=table[0][0]: return table[0][1:]
    for a,b in zip(table,table[1:]):
        if T<=b[0]:
            f=(T-a[0])/(b[0]-a[0]); return tuple(a[i]+f*(b[i]-a[i]) for i in (1,2))
    return table[-1][1:]


def sensible_enthalpy(T,T0):
    if T<T0:
        return tuple(-x for x in sensible_enthalpy(T0,T))
    knots=[T0]+[t for t in (26.85,326.85,626.85,776.85,926.85,1226.85,1426.85,1926.85,2476.85) if T0<t<T]+[T]
    return tuple(sum((heat_capacity(a)[i]+heat_capacity(b)[i])*(b-a)/2 for a,b in zip(knots,knots[1:])) for i in
    (0,1))


def main():
    ap=argparse.ArgumentParser(); ap.add_argument('action',choices=['initialize','prepare','accept','validate'])
    ap.add_argument('--root',type=Path,default=Path(__file__).resolve().parent)
    args=ap.parse_args(); root=args.root.resolve()
    c=load(root/'config.json'); validate_config(c); mesh=load(root/'mesh_map.json')
    if args.action=='validate':
        assert len(mesh['rows'])==c['geometry']['phenolic_radial_rows']
        assert all(len(r)==c['geometry']['faces_per_row'] for r in mesh['rows'])
        print('CONFIG_AND_MESH_VALID_NO_SOLVE'); return
    if args.action=='initialize':
        if (root/'state.json').exists(): raise RuntimeError('Existing run state preserved: use a new run
    directory.')
        atomic(root/'step_control.inp','')
        atomic(root/'accepted.inp','')
        save(root/'state.json',initial_state(mesh,c)); return
    s=load(root/'state.json')
    if s['config_sha256']!=fingerprint(c):
        raise ValueError('Configuration changed after initialization; create a new run, do not mix states.')
    if args.action=='prepare':
        import time as walltime, shutil
        control_root=Path(c.get('host_control_root',str(root.parents[1])))
        shared_runtime=Path(c.get('host_runtime',str(root)))
        guard_path = control_root / 'host_storage_status.json'
        guard = load(guard_path) if guard_path.exists() else {}
        storage_ok = (guard.get('allowed', False) and walltime.time()-guard.get('checked_epoch',0) < 90
                      and walltime.time()-guard.get('guest_checked_epoch',0) < 90
                      and guard.get('mac_free_bytes',0) >= 400_000_000_000
                      and shutil.disk_usage('C:\\').free >= 10_000_000_000)
        if not storage_ok:
            s['status']='PAUSED_STORAGE_GUARD'; save(root/'state.json',s)
            atomic(root/'step_control.inp','V0_OK=1\nV0_DONE=1\n'); return
        pause=(root/'PAUSE_REQUESTED').exists() or (shared_runtime/'PAUSE_REQUESTED').exists()
        stop=pause or s['status']=='CASE_PROPERTY_LIMIT_REACHED' or s['time_s']>=c['end_time_s']-1e-10 or s['
    layer']>=c['geometry']['phenolic_radial_rows']
        if stop:
            if pause:
                s['status']='PAUSED_AT_CHECKPOINT'; save(root/'state.json',s)
                if shared_runtime != root:save(shared_runtime/'state.json',s)
            atomic(root/'step_control.inp','V0_OK=1\nV0_DONE=1\n'); return
        s['status']='RUNNING'
        p,commands=prepare_step(mesh,c,s)
        save(root/'pending.json',p)
        atomic(root/'apply_loads.inp',commands)
        atomic(root/'step_control.inp',f'V0_OK=1\nV0_DONE=0\nV0_INDEX={p["index"]}\nV0_PREVIOUS={s["index"]}\
    nV0_TARGET={p["target"]:.16e}\nV0_DT={p["dt"]:.16e}\n')
    else:
        p=load(root/'pending.json')
        identity=[float(x) for x in (root/'solver_step_identity.txt').read_text().split()]
```

```
        validate_native_step_identity(identity,p)
        live=int(float((root/'live_topology_before_solve.txt').read_text().strip()))
        expected_live=len(mesh['elements'])-sum(len(r) for r in mesh['rows'][:p['layer']])
        if live!=expected_live: raise ValueError('Native pre-solve live topology mismatch')
        time=float((root/'observed_time.txt').read_text().strip())
        temp={k:v[0] for k,v in read_rows(root/'node_temperatures.txt').items()}
        new,record=accept_step(mesh,c,s,p,temp,read_rows(root/'element_fields.txt'),read_rows(root/'energy_fields.txt'),time)
        record['native_live_elements_before_solve']=live
        record['native_load_step']=int(identity[0]);record['native_substep']=int(identity[1])
        preserve_accepted_restart(root,new)
        # One immutable audit record per accepted state. CSV is derived from these.
        checkpoints=root/'checkpoints'; checkpoints.mkdir(exist_ok=True)
        # Compress state snapshots; audit records remain directly readable.
        import gzip
        dest=checkpoints/f'state_{new["index"]:06d}.json.gz'
        temporary=dest.with_suffix('.new')
        with gzip.open(temporary,'wt',encoding='utf-8') as stream: json.dump(new,stream,allow_nan=False)
        os.replace(temporary,dest)
        save(checkpoints/f'audit_{new["index"]:06d}.json',record)
        save(root/'state.json',new)
        # Guest-local native files avoid shared-drive latency. Mirror only
        # accepted compact state; full native outputs are copied at job exit.
        shared_runtime=Path(c.get('host_runtime',str(root)))
        if shared_runtime != root:
            import shutil
            shared_checkpoints=shared_runtime/'checkpoints'
            shared_checkpoints.mkdir(exist_ok=True)
            save(shared_checkpoints/f'audit_{new["index"]:06d}.json',record)
            shutil.copy2(dest,shared_checkpoints/dest.name)
            save(shared_runtime/'state.json',new)
        rows=[load(p) for p in sorted(checkpoints.glob('audit_*.json'))]
        import io
        buf=io.StringIO(); writer=csv.DictWriter(buf,fieldnames=list(record)); writer.writeheader(); writer.writerows(rows)
        atomic(root/c['output']['history_file'],buf.getvalue())
        atomic(root/'accepted.inp','V0_OK=1\n')


if __name__=='__main__':
    try: main()
    except Exception:
        # Missing success marker makes MAPDL stop before another SOLVE.
        trace=traceback.format_exc()
        print(trace)
        args_root=Path(sys.argv[sys.argv.index('--root')+1]) if '--root' in sys.argv else Path(__file__).resolve().parent
        atomic(args_root/'coupler_error.txt',trace)
        raise
```

A.8 Complete compiler and linker command file

The isolated build also contains an export definition with the two lines `EXPORTS` and `USERMATTH`. Compilation is not performed during report generation.

Source: `build_material.cmd`.

SHA-256: `2decff15d893d671b8fd49f79c24f53ba79d353b268ca1cd9435ad310ce928ea`.

```
@echo off
setlocal
rem Run in an isolated Iris build directory containing usermatth.F.
rem Compiler/link options follow the installed v261 ANSUSERSHARED.bat.
call "C:\Program Files (x86)\Intel\oneAPI\compiler\2023.1.0\env\vars.bat" intel64 vs2022
if errorlevel 1 exit /b 1
set "INCLUDE=C:\Program Files\ANSYS Inc\v261\ansys\customize\Include;%INCLUDE%"
set "LIB=C:\Program Files\ANSYS Inc\v261\ansys\custom\lib\winx64;%LIB%"
ifort /DNOSTDCALL /DARGTRAIL /DCADOE_ANSYS /DPCWINNT_SYS /D_EFL /DFORTRAN /O2 /MD /c /fpp /4Yportlib /auto /Fo.\
   /watch:source /DPCWIN64_SYS /DPCWINX64_SYS usermatth.F > compile.log 2>&1
if errorlevel 1 exit /b 1
link /out:usermatthLib.dll /def:usermatthLibex.def /dll /machine:X64 /map /manifest:embed /defaultlib:ANSYS.lib /
   defaultlib:ansMathUtils.lib /defaultlib:ansMemManager.lib /defaultlib:ansOpenMP.lib /defaultlib:ansUtils.lib
   usermatth.obj > link.log 2>&1
if errorlevel 1 exit /b 1
dumpbin /exports /dependents usermatthLib.dll > binary_inspection.log 2>&1
exit /b %errorlevel%
```

A.9 MAPDL-to-Python bridge

Source: `sustainer_al/couple.cmd`.
SHA-256: `b0aaa464ad046263e6059dfdf1604f299cd5d1e708503e6aa865577660ecb390`.

```
@echo off
"C:\Program Files\ANSYS Inc\v261\commonfiles\CPython\3_10\winx64\Release\python\python.exe" "C:\IrisStudy\runs\
   sustainer_al_pulse7_n80_midpoint_tight\coupler.py" %1 --root "C:\IrisStudy\runs\
   sustainer_al_pulse7_n80_midpoint_tight"
exit /b %ERRORLEVEL%
```

A.10 Complete representative MAPDL mesh and setup

Every material, node and element command is printed, without omitted connectivity. Dimensions are in metres and the initial temperature is in degrees Celsius. The conformal interface shares nodes; no hidden contact mesh is added.
Source: `sustainer_al/model_base.inp`.
SHA-256: `0b84b8ffe474819f1ec79e4b54bee820b6a4f39d6a22a701d7ae39fe55500aaf`.

```
/BATCH
/CONFIG,NOELDB,1
/TITLE,New motor phenolic consumption v0
/UNITS,MKS
/PREP7
SHPP,ON
ET,1,278
ET,2,278
KEYOPT,1,15,1
KEYOPT,2,15,1
! Phenolic UserMatTh: reference densities; fixed mesh; no thermal expansion.
TB,USER,1,10,15,THERM
TBTEMP,22.0
TBDATA,1,0.25,0.3,900.0,800.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,26.85
TBDATA,1,0.25,0.3,900.0,800.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,326.85
TBDATA,1,0.32,0.35,1100.0,900.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,626.85
TBDATA,1,0.36,0.4,1300.0,1000.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,776.85
TBDATA,1,0.36,0.44,1300.0,1250.0,1250.0,600.0
TBDATA,7,333,64081,1,418000.0,273.15,1600
TBDATA,13,300,1,1
TBTEMP,926.85
TBDATA,1,0.36,0.49,1300.0,1550.0,1250.0,600.0
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